

《随机过程》习题解答

习题 1

1. 令 $\{X(t), t \in T\}$ 为二阶矩存在的随机过程, 试证它是宽平稳的当且仅当 $E[X(s)]$ 与 $E[X(s)X(s+t)]$ 都不依赖于 s .

证 由宽平稳过程的定义知, $\{X(t), t \in T\}$ 宽平稳当且仅当下列两条件同时成立:

(1) $\mu_X(s) = E[X(s)]$ 在 $s \in T$ 上是常数;

(2) $R_X(t, s) = \text{Cov}[X(t), X(s)]$ 只与 $t - s$ 有关.

注意到

$$\text{Cov}[X(t), X(s)] = E[X(t)X(s)] - E[X(t)]E[X(s)],$$

因此, 条件 (1) 和 (2) 与下列两条件等价:

(3) $\mu_X(s) = E[X(s)]$ 在 $s \in T$ 上是常数;

(4) $E[X(t)X(s)]$ 只与 $t - s$ 有关.

这就证明了 $\{X(t), t \in T\}$ 宽平稳当且仅当 $E[X(s)]$ 与 $E[X(s)X(s+t)]$ 都不依赖于 s . ■

2. 记 $U_1, \dots, U_n$ 为在 $(0, 1)$ 中均匀分布的 n 个独立随机变量. 对 $0 < t, x < 1$, 定义

$$I(t, x) = \begin{cases} 1, & x \leq t, \\ 0, & x > t, \end{cases}$$

并记 $X(t) = \frac{1}{n} \sum_{k=1}^n I(t, U_k)$, $0 \leq t \leq 1$, 这是 $U_1, \dots, U_n$ 的经验分布函数. 试求随机过程 $\{X(t), 0 \leq t \leq 1\}$ 的均值和自协方差函数.

解 由题设知, $\{X(t), 0 \leq t \leq 1\}$ 的均值函数为

$$\begin{aligned} \mu_X(t) &= E[X(t)] = \frac{1}{n} \sum_{k=1}^n E[I(t, U_k)] \\ &= E[I(t, U_1)], \quad 0 \leq t \leq 1, \end{aligned} \quad (1)$$

自协方差函数为

$$\begin{aligned} R_X(t, s) &= \text{Cov}[X(t), X(s)] \\ &= \frac{1}{n^2} \sum_{k=1}^n \sum_{l=1}^n \text{Cov}[I(t, U_k), I(s, U_l)] \\ &= \frac{1}{n^2} \sum_{k=1}^n \text{Cov}[I(t, U_k), I(s, U_k)] \\ &= \frac{1}{n} \text{Cov}[I(t, U_1), I(s, U_1)], \quad 0 \leq t \leq 1, \end{aligned} \quad (2)$$

而

$$E[I(t, U_1)] = P(U_1 \leq t) = t, \quad 0 \leq t \leq 1, \quad (3)$$

$$\begin{aligned}
 E[I(t, U_1)I(s, U_1)] &= P(U_1 \leq t, U_1 \leq s) \\
 &= P(U_1 \leq \min\{t, s\}) = \min\{t, s\}, \quad 0 \leq t, s \leq 1, \\
 Cov[I(t, U_1), I(s, U_1)] &= E[I(t, U_1)I(s, U_1)] - E[I(t, U_1)]E[I(s, U_1)] \\
 &= \min\{t, s\} - ts, \quad 0 \leq t, s \leq 1,
 \end{aligned} \tag{4}$$

将 (3) 和 (4) 代入 (1) 和 (2) 中得

$$\begin{aligned}
 \mu_X(t) &= t, \quad 0 \leq t \leq 1, \\
 R_X(t, s) &= \frac{1}{n}[\min\{t, s\} - ts], \quad 0 \leq t, s \leq 1.
 \end{aligned}$$

■

3. 令 Z_1, Z_2 为独立同分布的正态随机变量, 均值为 0, 方差为 σ^2 , λ 为实数. 定义 $X(t) = Z_1 \cos(\lambda t) + Z_2 \sin(\lambda t)$. 试求 $\{X(t), t \in (-\infty, +\infty)\}$ 的均值函数和自协方差函数. 它是宽平稳的吗?

解 由题设可知, $\{X(t), t \in (-\infty, +\infty)\}$ 的均值函数为

$$\mu_X(t) = E(Z_1) \cos(\lambda t) + E(Z_2) \sin(\lambda t) = 0, \quad t \in (-\infty, +\infty),$$

自协方差函数为

$$\begin{aligned}
 R_X(t, s) &= Cov[Z_1 \cos(\lambda t) + Z_2 \sin(\lambda t), Z_1 \cos(\lambda s) + Z_2 \sin(\lambda s)] \\
 &= Cov(Z_1, Z_1) \cos(\lambda t) \cos(\lambda s) + Cov(Z_1, Z_2) \cos(\lambda t) \sin(\lambda s) \\
 &\quad + Cov(Z_2, Z_1) \sin(\lambda t) \cos(\lambda s) + Cov(Z_2, Z_2) \sin(\lambda t) \sin(\lambda s) \\
 &= \sigma^2 \cos(\lambda t) \cos(\lambda s) + \sigma^2 \sin(\lambda t) \sin(\lambda s) \\
 &= \sigma^2 \cos[\lambda(t - s)], \quad t, s \in (-\infty, +\infty).
 \end{aligned}$$

故 $\{X(t), t \in (-\infty, +\infty)\}$ 是宽平稳的. ■

4. Poisson 过程 $\{X(t), t \geq 0\}$ 满足 (i) $X(0) = 0$; (ii) 对 $t > s$, $X(t) - X(s)$ 服从均值为 $\lambda(t - s)$ 的 Poisson 分布; (iii) 过程是有独立增量的. 试求其均值函数和自协方差函数. 它是宽平稳的吗?

解 $\{X(t), t \geq 0\}$ 的均值函数为

$$\begin{aligned}
 \mu_X(t) &= E[X(t)] = E[X(t) - X(0)] \\
 &= \lambda t, \quad t > 0,
 \end{aligned}$$

$\{X(t), t \geq 0\}$ 的方差函数为

$$\begin{aligned}
 Var[X(t)] &= Var[X(t) - X(0)] \\
 &= \lambda t, \quad t > 0,
 \end{aligned}$$

对 $s > t > 0$, 有

$$E[X(t)X(s)] = E\{[X(t) - X(0)][(X(s) - X(t)) + (X(t) - X(0))]\}$$

$$\begin{aligned}
&= E\{[X(t) - X(0)]^2\} + E\{[X(t) - X(0)][X(s) - X(t)]\} \\
&= \text{Var}[X(t) - X(0)] + \{E[X(t) - X(0)]\}^2 \\
&\quad + E[X(t) - X(0)]E[X(s) - X(t)] \\
&= \lambda t + (\lambda t)^2 + \lambda t \cdot \lambda(s - t) \\
&= \lambda t(\lambda s + 1),
\end{aligned}$$

因此, $\{X(t), t \geq 0\}$ 的自协方差函数为

$$\begin{aligned}
R_X(t, s) &= E[X(t)X(s)] - E[X(t)]E[X(s)] \\
&= \lambda t, \quad s \geq t > 0,
\end{aligned}$$

自相关函数为

$$\begin{aligned}
r_X(t, s) &= \frac{R_X(t, s)}{[R_X(t, t)R_X(s, s)]^{1/2}} \\
&= \sqrt{\frac{t}{s}}, \quad s \geq t > 0.
\end{aligned}$$

■

5. $\{X(t), t \geq 0\}$ 为第四题中的 Poisson 过程. 记 $Y(t) = X(t+1) - X(t)$, 试求过程 $\{Y(t), t \geq 0\}$ 的均值函数和自协方差函数, 并研究其平稳性.

解 $\{Y(t), t \geq 0\}$ 的均值函数为

$$\begin{aligned}
\mu_Y(t) &= E[X(t+1)] - E[X(t)] = \mu_X(t+1) - \mu_X(t) \\
&= \lambda, \quad t \geq 0,
\end{aligned}$$

自协方差函数为

$$\begin{aligned}
R_Y(t, s) &= \text{Cov}[X(t+1) - X(t), X(s+1) - X(s)] \\
&= \text{Cov}[X(t+1), X(s+1)] - \text{Cov}[X(t+1), X(s)] \\
&\quad - \text{Cov}[X(t), X(s+1)] + \text{Cov}[X(t), X(s)] \\
&= \lambda(\min\{t, s\} + 1) - \lambda \min\{t+1, s\} - \lambda \min\{t, s+1\} + \lambda \min\{t, s\}, \\
&= \begin{cases} 0, & \text{当 } 0 \leq t < s-1, \\ \lambda(t-s+1), & \text{当 } s-1 \leq t < s, \\ \lambda(s-t+1), & \text{当 } s \leq t < s+1, \\ 0, & \text{当 } t \geq s+1, \end{cases} \quad t, s \geq 0.
\end{aligned}$$

这说明了 $\{Y(t), t \geq 0\}$ 是宽平稳的. ■

6. 令 Z_1 和 Z_2 是独立同分布的随机变量, $P(Z_1 = -1) = P(Z_1 = 1) = \frac{1}{2}$. 记 $X(t) = Z_1 \cos(\lambda t) + Z_2 \sin(\lambda t), t \in R$, 试证 $\{X(t), t \in R\}$ 是宽平稳的, 它是严平稳的吗?

证明 由题设知, $\{X(t), t \in R\}$ 的均值函数为

$$\mu_X(t) = E(Z_1) \cos(\lambda t) + E(Z_2) \sin(\lambda t) = 0, \quad t \in R, \quad (1)$$

自协方差函数为

$$\begin{aligned}
 R_X(t, s) &= \text{Cov}(Z_1 \cos(\lambda t) + Z_2 \sin(\lambda t), Z_1 \cos(\lambda s) + Z_2 \sin(\lambda s)) \\
 &= \text{Var}(Z_1) \cos(\lambda t) \cos(\lambda s) + \text{Var}(Z_2) \sin(\lambda t) \sin(\lambda s) \\
 &= \cos(\lambda t) \cos(\lambda s) + \sin(\lambda t) \sin(\lambda s) \\
 &= \cos(\lambda(t - s)), \quad t, s \in R.
 \end{aligned} \tag{2}$$

由 (1) 和 (2) 即知, $\{X(t), t \in R\}$ 是宽平稳的. 进而, 由题设可知, 随机变量 $X(t)$ 的矩母函数为

$$\begin{aligned}
 g_{X(t)}(u) &= E(e^{uX(t)}) = E\{\exp[u(Z_1 \cos(\lambda t) + Z_2 \sin(\lambda t))]\} \\
 &= E\{\exp[uZ_1 \cos(\lambda t)]\} \cdot E\{\exp[uZ_2 \sin(\lambda t)]\} \\
 &= \frac{1}{4} \{\exp[-u \cos(\lambda t)] + \exp[u \cos(\lambda t)]\} \\
 &\quad \cdot \{\exp[-u \sin(\lambda t)] + \exp[u \sin(\lambda t)]\}, \quad u \in R,
 \end{aligned}$$

这说明了 $X(t)$ 的分布与 $t \in R$ 有关, 因此 $\{X(t), t \in R\}$ 不是严平稳的. ■

7. 试证: 若 $Z_0, Z_1, Z_2, \dots$ 为独立同分布随机变量序列, 定义 $X(n) = Z_0 + Z_1 + \dots + Z_n, n = 0, 1, 2, \dots$, 则 $\{X(n), n = 0, 1, 2, \dots\}$ 是独立增量过程.

证明 注意到对任意 n 及任意 $t_1, \dots, t_n \in \{0, 1, 2, \dots\}, t_1 < t_2 < \dots < t_n$, 有

$$\begin{cases} X(t_2) - X(t_1) = Z_{t_1+1} + \dots + Z_{t_2}, \\ X(t_3) - X(t_2) = Z_{t_2+1} + \dots + Z_{t_3}, \\ \dots\dots\dots \\ X(t_n) - X(t_{n-1}) = Z_{t_{n-1}+1} + \dots + Z_{t_n}. \end{cases} \tag{1}$$

而由题设知, $Z_{t_1+1}, \dots, Z_{t_n}$ 互相独立, 因此 $(Z_{t_1+1}, \dots, Z_{t_2}), (Z_{t_2+1}, \dots, Z_{t_3}), \dots, (Z_{t_{n-1}+1}, \dots, Z_{t_n})$ 互相独立, 故由 (1) 知, $X(t_2) - X(t_1), X(t_3) - X(t_2), \dots, X(t_n) - X(t_{n-1})$ 互相独立. 这就证明了 $\{X(n), n = 0, 1, 2, \dots\}$ 是独立增量过程. ■

8. 若 $X_1, X_2, \dots$ 是一列独立随机变量, 还要添加什么条件才能确保 $\{X_1, X_2, \dots\}$ 是严平稳的随机过程?

解 若 $\{X_1, X_2, \dots\}$ 严平稳, 则对任意正整数 m 和 n , X_m 和 X_n 的分布都相同, 从而 $X_1, X_2, \dots$ 是一列同分布的随机变量. 而当 $X_1, X_2, \dots$ 是一列独立同分布的随机变量时, 对任意正整数 k 及 $n_1, \dots, n_k, k$ 维随机向量 $(X_{n_1}, \dots, X_{n_k})$ 的分布函数为 (记 $X_1, X_2, \dots$ 共同的分布函数为 $F(x)$)

$$\begin{aligned}
 F_{(X_{n_1}, \dots, X_{n_k})}(x_1, \dots, x_k) &= F_{X_{n_1}}(x_1) \cdots F_{X_{n_k}}(x_k) \\
 &= F(x_1) \cdots F(x_k), \quad -\infty < x_1, \dots, x_k < +\infty,
 \end{aligned}$$

这说明了 $(X_{n_1}, \dots, X_{n_k})$ 的分布函数与 $n_1, \dots, n_k$ 无关, 故 $\{X_1, X_2, \dots\}$ 严平稳. ■

9. 令 X 和 Y 是从单位圆内的均匀分布中随机选取一点所得的横坐标和纵坐标. 试计算条件概率

$$P\left(X^2 + Y^2 \geq \frac{3}{4} \mid X > Y\right).$$

解 注意到 (X, Y) 的联合密度函数为

$$f(x, y) = \begin{cases} \frac{1}{\pi}, & x^2 + y^2 \leq 1. \\ 0, & \text{其他.} \end{cases}$$

因此

$$\begin{aligned} P(X > Y) &= \iint_{x>y} f(x, y) dx dy \\ &= \frac{1}{\pi} \iint_{x^2+y^2 \leq 1, x>y} dx dy \\ &= \frac{1}{2}, \\ P(X^2 + Y^2 \geq \frac{3}{4}, X > Y) &= \iint_{x^2+y^2 \geq \frac{3}{4}, x>y} f(x, y) dx dy \\ &= \frac{1}{\pi} \iint_{\frac{3}{4} \leq x^2+y^2 \leq 1, x>y} dx dy \\ &= \frac{1}{2} \left[1 - \left(\frac{3}{4} \right)^2 \right] = \frac{7}{32}, \end{aligned}$$

故所求条件概率为

$$P\left(X^2 + Y^2 \geq \frac{3}{4} \mid X > Y\right) = \frac{P(X^2 + Y^2 \geq \frac{3}{4}, X > Y)}{P(X > Y)} = \frac{7}{16}.$$

■

10. 离子依参数为 λ 的 Poisson 分布进入计数器, 两离子到达的时间间隔 $T_1, T_2, \dots$ 是独立的参数为 λ 的指数分布随机变量. 记 S 是 $[0, 1]$ 时段中的离子总数, 时间区间 $I \subset [0, 1]$, 其长度记为 $|I|$. 试证明 $P(T_1 \in I, S = 1) = P(T_1 \in I, T_1 + T_2 > 1)$, 并由此计算 $P(T_1 \in I \mid S = 1) = |I|$.

解证 由题设可知, 第 i 个离子在 $T_1 + \dots + T_i$ 时刻进入计数器, $i = 1, 2, \dots$, 而 S 是 $[0, 1]$ 时段内进入计数器的离子总数, 因此

$$\{T_1 \in I, S = 1\} = \{T_1 \in I, T_1 + T_2 > 1\},$$

故

$$P(T_1 \in I, S = 1) = P(T_1 \in I, T_1 + T_2 > 1). \quad (1)$$

而 (T_1, T_2) 的联合密度函数为

$$f_{(T_1, T_2)}(t_1, t_2) = f_{T_1}(t_1)f_{T_2}(t_2) = \lambda^2 e^{-\lambda(t_1+t_2)}, \quad t_1, t_2 > 0,$$

因而

$$\begin{aligned} P(T_1 \in I, T_1 + T_2 > 1) &= \iint_{t_1 \in I, t_1+t_2 > 1} f_{(T_1, T_2)}(t_1, t_2) dt_1 dt_2 \\ &= \lambda^2 \iint_{t_1, t_2 > 0, t_1 \in I, t_1+t_2 > 1} e^{-\lambda(t_1+t_2)} dt_1 dt_2. \end{aligned} \quad (2)$$

作变换

$$u = t_1, \quad v = t_1 + t_2,$$

则可将 (2) 中积分表为

$$\begin{aligned}
 P(T_1 \in I, T_1 + T_2 > 1) &= \lambda^2 \iint_{u \in I, v > 1} e^{-\lambda v} du dv \\
 &= \lambda^2 \int_{u \in I} du \int_1^{+\infty} e^{-\lambda v} dv \\
 &= \lambda |I| e^{-\lambda}.
 \end{aligned} \tag{3}$$

而

$$P(S = 1) = \lambda e^{-\lambda},$$

由此及 (1) 和 (3) 可得

$$P(T_1 \in I | S = 1) = \frac{P(T_1 \in I, S = 1)}{P(S = 1)} = |I|.$$

■

12. 气体分子的速度 V 有三个垂直分量 V_x, V_y, V_z , 它们的联合密度函数依 Maxwell-Boltzman 定律为

$$f_{V_x, V_y, V_z}(v_x, v_y, v_z) = \frac{1}{(2\pi kT)^{3/2}} \exp \left\{ -\frac{v_x^2 + v_y^2 + v_z^2}{2kT} \right\}, \quad -\infty < v_x, v_y, v_z < +\infty,$$

其中 K 是 Boltzman 常数, T 是绝对温度, 给定分子的总动能为 e . 试求 x 方向的动量的绝对值的期望值.

解 由于 V_x, V_y, V_z 的联合密度函数为

$$\begin{aligned}
 f_{V_x, V_y, V_z}(v_x, v_y, v_z) &= \frac{1}{(2\pi kT)^{3/2}} \exp \left\{ -\frac{v_x^2 + v_y^2 + v_z^2}{2kT} \right\} \\
 &= \frac{1}{(2\pi kT)^{1/2}} \exp \left\{ -\frac{v_x^2}{2kT} \right\} \cdot \frac{1}{(2\pi kT)^{1/2}} \exp \left\{ -\frac{v_y^2}{2kT} \right\} \\
 &\quad \cdot \frac{1}{(2\pi kT)^{1/2}} \exp \left\{ -\frac{v_z^2}{2kT} \right\}, \quad -\infty < v_x, v_y, v_z < +\infty,
 \end{aligned}$$

因此, V_x, V_y, V_z 互相独立, 且 V_x, V_y, V_z 都服从正态分布 $N(0, kT)$. 故气体分子的总动能为

$$e = \frac{1}{2} m E(V_x^2 + V_y^2 + V_z^2) = \frac{3}{2} m k T,$$

由此可得

$$m = \frac{2e}{3kT}, \tag{1}$$

而气体 x 方向的动量的绝对值的期望值为

$$\begin{aligned}
 m E(|V_x|) &= \frac{m}{(2\pi kT)^{1/2}} \int_{-\infty}^{+\infty} |v_x| \exp \left\{ -\frac{v_x^2}{2kT} \right\} dv_x \\
 &= \frac{m}{(2\pi kT)^{1/2}} \left[- \int_{-\infty}^0 v_x \exp \left\{ -\frac{v_x^2}{2kT} \right\} dv_x \right.
 \end{aligned}$$

$$\begin{aligned}
& + \int_0^{+\infty} v_x \exp \left\{ -\frac{v_x^2}{2kT} \right\} dv_x \Big] \\
& = \frac{2m}{(2\pi kT)^{1/2}} \int_0^{+\infty} v_x \exp \left\{ -\frac{v_x^2}{2kT} \right\} dv_x \\
& = \frac{m}{(2\pi kT)^{1/2}} \int_0^{+\infty} \exp \left\{ -\frac{v_x^2}{2kT} \right\} d(v_x^2) \\
& = m \left(\frac{2kT}{\pi} \right)^{1/2}.
\end{aligned}$$

由此及 (1) 可得

$$mE(|V_x|) = \frac{2e}{3} \left(\frac{2}{\pi kT} \right)^{1/2}.$$

■

13. 若随机变量 $X_1, \dots, X_n$ 独立同分布, 分布是参数为 λ 的指数分布. 试证 $T = \sum_{i=1}^n X_i$ 服从参数为 (n, λ) 的 Γ 分布, 其密度为

$$f(t) = \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t}, \quad t \geq 0.$$

证明 注意到 $T = \sum_{i=1}^n X_i$ 的矩母函数为

$$g_T(t) = [g_{X_1}(t)]^n, \quad (1)$$

其中 $g_{X_1}(t)$ 为 X_1 的矩母函数. 由于 X_1 服从参数为 λ 的指数分布, 因此

$$g_{X_1}(t) = \lambda \int_0^{+\infty} e^{-(\lambda-t)x} dx = \frac{\lambda}{\lambda-t}, \quad t < \lambda,$$

将其代入 (1) 中得

$$g_T(t) = \left(\frac{\lambda}{\lambda-t} \right)^n, \quad t < \lambda. \quad (2)$$

又参数为 (n, λ) 的 Γ 分布的矩母函数为

$$\begin{aligned}
g(t) &= \frac{\lambda^n}{(n-1)!} \int_0^{+\infty} x^{n-1} e^{-(\lambda-t)x} dx \\
&= \frac{\lambda^n}{(n-1)!} \frac{(n-1)!}{(\lambda-t)^n} \\
&= \left(\frac{\lambda}{\lambda-t} \right)^n, \quad t < \lambda.
\end{aligned}$$

由此及 (2) 即知 $T = \sum_{i=1}^n X_i$ 服从参数为 (n, λ) 的 Γ 分布. ■

14. 设 X_1 和 X_2 分别为相互独立的均值为 λ_1 和 λ_2 的 Poisson 随机变量. 试求 $X_1 + X_2$ 的分布, 并计算给定 $X_1 + X_2 = n$ 时, X_1 的条件分布.

解 注意到若 $X \sim P(\lambda)$, 则 X 的矩母函数为

$$\begin{aligned} g_X(t) &= E(e^{tX}) = \sum_{i=0}^{+\infty} e^{it} \frac{\lambda^i}{i!} e^{-\lambda} \\ &= e^{-\lambda} \sum_{i=0}^{+\infty} \frac{(\lambda e^t)^i}{i!} = \exp\{\lambda(e^t - 1)\}, \quad t \in (-\infty, +\infty), \end{aligned}$$

因此, $X_1 + X_2$ 的矩母函数为

$$\begin{aligned} g_{X_1+X_2}(t) &= g_{X_1}(t)g_{X_2}(t) \\ &= \exp\{\lambda_1(e^t - 1)\} \exp\{\lambda_2(e^t - 1)\} \\ &= \exp\{(\lambda_1 + \lambda_2)(e^t - 1)\}, \quad t \in (-\infty, +\infty), \end{aligned}$$

这说明了 $X_1 + X_2 \sim P(\lambda_1 + \lambda_2)$. 故

$$P(X_1 + X_2 = n) = \frac{(\lambda_1 + \lambda_2)^n}{n!} e^{-(\lambda_1 + \lambda_2)}, \quad n = 0, 1, 2, \dots \quad (1)$$

进而有

$$\begin{aligned} P(X_1 + X_2 = n, X_1 = m) &= P(X_1 = m, X_2 = n - m) = P(X_1 = m)P(X_2 = n - m) \\ &= \frac{\lambda_1^m}{m!} e^{-\lambda_1} \cdot \frac{\lambda_2^{n-m}}{(n-m)!} e^{-\lambda_2}, \quad m, n = 0, 1, 2, \dots, m \leq n, \end{aligned}$$

由此及 (1) 可得

$$\begin{aligned} P(X_1 = m \mid X_1 + X_2 = n) &= \frac{P(X_1 = m, X_1 + X_2 = n)}{P(X_1 + X_2 = n)} \\ &= \binom{n}{m} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^m \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-m}, \\ &\quad m, n = 0, 1, 2, \dots, m \leq n. \end{aligned}$$

■

15. 若 $X_1, X_2, \dots$ 独立且有相同的以 λ 为参数的指数分布, N 与 $X_1, X_2, \dots$ 独立, N 服从几何分布

$$P(N = n) = \beta(1 - \beta)^{n-1}, \quad n = 1, 2, \dots, 0 < \beta < 1.$$

试求随机和 $\sum_{i=1}^N X_i$ 的分布.

解 由题设可知, 已知 $N = n$ 时, $X_1, X_2, \dots$ 独立且有相同的以 λ 为参数的指数分布, 因此由指数分布的可加性知, 已知 $N = n$ 时, $\sum_{i=1}^n X_i$ 服从以 (n, λ) 为参数的 Γ 分布

$$f_{Y|N}(y \mid n) = \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t}, \quad t \geq 0.$$

故随机和 $Y = \sum_{i=1}^N X_i$ 的分布为

$$f_Y(y) = \sum_{n=1}^{\infty} f_{Y|N}(y \mid n) P(N = n)$$

$$\begin{aligned}
&= \lambda \beta e^{-\lambda t} \sum_{n=1}^{\infty} \frac{[\lambda(1-\beta)t]^{n-1}}{(n-1)!} \\
&= \lambda \beta e^{-\lambda t} e^{\lambda(1-\beta)t} \\
&= \lambda \beta e^{-\lambda \beta t}, \quad t \geq 0.
\end{aligned}$$

这说明了 Y 服从参数为 $\lambda\beta$ 的指数分布. ■

16. 若 $X_1, X_2, \dots$ 独立同分布, $P(X_1 = \pm 1) = \frac{1}{2}$, N 与 $X_i, i = 1, 2, \dots$ 独立且服从参数为 β 的几何分布, $0 < \beta < 1$. 试求随机和 $Y = \sum_{i=1}^N X_i$ 的均值, 方差和三、四阶矩.

解 1 由题设可知, 对任意正整数 n , 有

$$\begin{aligned}
E(Y \mid N = n) &= E\left(\sum_{i=1}^N X_i \mid N = n\right) = E\left(\sum_{i=1}^n X_i \mid N = n\right) \\
&= E\left(\sum_{i=1}^n X_i\right) = nE(X_1) = 0,
\end{aligned}$$

$$\begin{aligned}
E(Y^2 \mid N = n) &= E\left[\left(\sum_{i=1}^N X_i\right)^2 \mid N = n\right] = E\left[\left(\sum_{i=1}^n X_i\right)^2 \mid N = n\right] \\
&= E\left[\left(\sum_{i=1}^n X_i\right)^2\right] = \sum_{j=1}^n \sum_{i=1}^n E(X_i X_j) \\
&= \sum_{i=1}^n E(X_i^2) = n,
\end{aligned}$$

$$\begin{aligned}
E(Y^3 \mid N = n) &= E\left[\left(\sum_{i=1}^N X_i\right)^3 \mid N = n\right] = E\left[\left(\sum_{i=1}^n X_i\right)^3 \mid N = n\right] \\
&= E\left[\left(\sum_{i=1}^n X_i\right)^3\right] = \sum_{j=1}^n \sum_{i=1}^n \sum_{k=1}^n E(X_i X_j X_k) = 0,
\end{aligned}$$

$$\begin{aligned}
E(Y^4 \mid N = n) &= E\left[\left(\sum_{i=1}^N X_i\right)^4 \mid N = n\right] = E\left[\left(\sum_{i=1}^n X_i\right)^4 \mid N = n\right] \\
&= E\left[\left(\sum_{i=1}^n X_i\right)^4\right] = \sum_{j=1}^n \sum_{i=1}^n \sum_{k=1}^n \sum_{l=1}^n E(X_i X_j X_k X_l) \\
&= \sum_{i=1}^n E(X_i^4) + \sum_{i,j=1, i \neq j}^n E(X_i^2 X_j^2) = n^2,
\end{aligned}$$

故 Y 的均值为

$$E(Y) = E[E(Y \mid N)] = 0,$$

二阶矩为

$$E(Y^2) = E[E(Y^2 \mid N)] = E(N)$$

$$= \beta \sum_{n=1}^{\infty} n(1-\beta)^{n-1} = \frac{1}{\beta},$$

三阶矩为

$$E(Y^3) = E[E(Y^3 | N)] = 0,$$

四阶矩为

$$\begin{aligned} E(Y^4) &= E[E(Y^4 | N)] = E(N^2) \\ &= \beta \sum_{n=1}^{\infty} n^2(1-\beta)^{n-1} = \frac{1}{\beta}. \end{aligned}$$

解 2 由题设可知, 对任意正整数 n , 有

$$\begin{aligned} E(e^{tY} | N = n) &= E(\exp\{t \sum_{i=1}^N X_i\} | N = n) = E(\exp\{t \sum_{i=1}^n X_i\} | N = n) \\ &= E(\exp\{t \sum_{i=1}^n X_i\}) = [E(e^{tX_1})]^n \\ &= \left[\frac{1}{2}(e^{-t} + e^t) \right]^n, \end{aligned}$$

因此

$$E(e^{tY} | N) = \left[\frac{1}{2}(e^{-t} + e^t) \right]^N,$$

由此可得 Y 的矩母函数为

$$\begin{aligned} g_Y(t) &= E(e^{tY}) = E[E(e^{tY} | N)] \\ &= \beta \sum_{n=1}^{\infty} \left[\frac{1}{2}(e^{-t} + e^t) \right]^n (1-\beta)^{n-1} \\ &= \frac{\beta(e^{-t} + e^t)}{2 - (1-\beta)(e^{-t} + e^t)}. \end{aligned}$$

故 Y 的均值为

$$E(Y) = \frac{dg_Y(t)}{dt} \Big|_{t=0} = 0,$$

二阶矩为

$$E(Y^2) = \frac{d^2 g_Y(t)}{dt^2} \Big|_{t=0} = \frac{1}{\beta},$$

三阶矩为

$$E(Y^3) = \frac{d^3 g_Y(t)}{dt^3} \Big|_{t=0} = 0,$$

四阶矩为

$$E(Y^4) = \frac{d^4 g_Y(t)}{dt^4} \Big|_{t=0} = \frac{1}{\beta}.$$

■

17. 随机变量 N 服从参数为 λ 的 Poisson 分布. 给定 $N = n$, 随机变量 M 服从以 n 和 p 为参数的二项分布. 试求 M 的无条件概率分布.

解 由题设可知

$$P(M = m | N = n) = \binom{n}{m} p^m (1-p)^{n-m}, \quad m = 0, 1, 2, \dots, n, n = 0, 1, 2, \dots,$$

$$P(N = n) = \frac{\lambda^n e^{-\lambda}}{n!}, \quad n = 0, 1, 2, \dots,$$

故 M 的无条件概率分布为

$$\begin{aligned} P(M = m) &= \sum_{n=m}^{\infty} P(M = m | N = n) P(N = n) \\ &= p^m e^{-\lambda} \sum_{n=m}^{\infty} \binom{n}{m} \frac{(1-p)^{n-m} \lambda^n}{n!} \\ &= \frac{(\lambda p)^m e^{-\lambda}}{m!} \sum_{n=0}^{\infty} \frac{[(1-p)\lambda]^n}{n!} \\ &= \frac{(\lambda p)^m e^{-\lambda p}}{m!}, \quad m = 0, 1, 2, \dots \end{aligned}$$

这说明了 Y 服从参数为 λp 的 Poisson 分布. ■

习题 2

1. 设 $\{N(t) : t \geq 0\}$ 是一强度为 λ 的 Poisson 过程, 对 $0 \leq s < t$, 试求条件概率 $P(N(s) = k | N(t) = n)$.

解 由题设可知

$$P(N(t) = n) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, \quad n = 0, 1, 2, \dots, t > 0.$$

$$\begin{aligned} P(N(s) = k, N(t) = n) &= P(N(s) = k, N(t) - N(s) = n - k) \\ &= P(N(s) = k)P(N(t) - N(s) = n - k) \\ &= \frac{(\lambda s)^k}{k!} e^{-\lambda s} \cdot \frac{[\lambda(t-s)]^{n-k}}{(n-k)!} e^{-\lambda(t-s)} \\ &= \frac{\lambda^n s^k (t-s)^{n-k}}{k!(n-k)!} e^{-\lambda t}, \\ &\quad k = 0, 1, \dots, n, n = 0, 1, 2, \dots, 0 \leq s < t. \end{aligned}$$

故

$$\begin{aligned} P(N(s) = k | N(t) = n) &= \frac{P(N(s) = k, N(t) = n)}{P(N(t) = n)} \\ &= \frac{n!}{k!(n-k)!} \left(\frac{s}{t}\right)^k \left(1 - \frac{s}{t}\right)^{n-k}, \\ &\quad k = 0, 1, \dots, n, n = 0, 1, 2, \dots, 0 \leq s < t. \end{aligned}$$

这说明了, $N(s) | N(t) = n \sim B(n, \frac{s}{t}), 0 \leq s < t$. ■

2. 设 $\{N(t) : t \geq 0\}$ 是一强度为 λ 的 Poisson 过程, 对 $t > 0, s \geq s$, 试计算 $E[N(t)N(t+s)]$.

解 由题设可知

$$\begin{aligned} E[N(t)N(t+s)] &= E\{N(t)[(N(t+s) - N(t)) + N(t)]\} \\ &= E[N(t)(N(t+s) - N(t))] + E[N(t)^2] \\ &= E[N(t)]E[N(t+s) - N(t)] + \text{Var}[N(t)] + [E(N(t))]^2 \\ &= \lambda t \cdot \lambda s + \lambda t + (\lambda t)^2 \\ &= \lambda t[\lambda(t+s) + 1]. \end{aligned}$$

■

3. 电报依平均速率为每小时 3 个的 Poisson 过程达到电报局, 试问

- (i) 从早上 8 时到中午没收到电报的概率;
- (ii) 下午第一份电报达到时间的分布是什么?

解 从早上 8 时开始计时, 以小时为计时单位, 则

(i) 所求概率为

$$P(N(4) = 0) = e^{-3 \cdot 4} \approx 6.1442 \times 10^{-6}.$$

(ii) 记下午第一份电报达到时间为 T_1 , 则

$$P(T_1 > t) = P(N(t) = N(4)) = e^{-3(t-4)}, \quad t > 4.$$

由此可得 T_1 的分布函数为

$$F_{T_1}(t) = 1 - P(T_1 > t) = 1 - e^{-3(t-4)}, \quad t > 4,$$

密度函数为

$$f_{T_1}(t) = \frac{dF_{T_1}(t)}{dt} = 3e^{-3(t-4)}, \quad t > 4.$$

这说明了 $T_1 - 4$ 服从参数为 3 的指数分布. ■

4. $\{N(t) : t \geq 0\}$ 为一 $\lambda = 2$ 的 Poisson 过程, 试求

(i) $P(N(1) \leq 2)$;

(ii) $P(N(1) = 1 \text{ 且 } N(2) = 3)$;

(iii) $P(N(1) \geq 2 \mid N(1) \geq 1)$.

解 (i)

$$\begin{aligned} P(N(1) \leq 2) &= P(N(1) = 0) + P(N(1) = 1) + P(N(1) = 2) \\ &= e^{-2} + \frac{2^1}{1!}e^{-2} + \frac{2^2}{2!}e^{-2} \\ &\approx 0.6767. \end{aligned}$$

(ii)

$$\begin{aligned} P(N(1) = 1 \text{ 且 } N(2) = 3) &= P(N(1) = 1, N(2) - N(1) = 2) \\ &= P(N(1) = 1)P(N(2) - N(1) = 2) \\ &= \frac{2^1}{1!}e^{-2} \cdot \frac{[2(2-1)]^2}{2!}e^{-2(2-1)} \\ &\approx 0.0733. \end{aligned}$$

(iii)

$$\begin{aligned} P(N(1) \geq 2 \mid N(1) \geq 1) &= \frac{P(N(1) \geq 2, N(1) \geq 1)}{P(N(1) \geq 1)} \\ &= \frac{P(N(1) \geq 2)}{P(N(1) \geq 1)} = \frac{1 - P(N(1) = 0) - P(N(1) = 1)}{1 - P(N(1) = 0)} \\ &= 1 - \frac{P(N(1) = 1)}{1 - P(N(1) = 0)} = 1 - \frac{\frac{2^1}{1!}e^{-2}}{1 - e^{-2}} \\ &\approx 0.6870. \end{aligned}$$

6. 一部 600 页的著作总共有 240 个印刷错误, 试利用 Poisson 过程近似求出某连续 3 页有 k 个印刷错误的概率, $k = 0, 1, 2, \dots, 240$.

解 所求概率为

$$P(N(m+3) - N(m) = k \mid N(600) = 240),$$

$$k = 0, 1, 2, \dots, 240, m = 0, 1, 2, \dots, 597. \quad (1)$$

由于

$$\begin{aligned} & P(N(m+3) - N(m) = k, N(600) = 240) \\ &= \sum_{l=0}^{240-k} P(N(m) = l, N(m+3) - N(m) = k, N(600) - N(m+3) = 240 - k - l) \\ &= \sum_{l=0}^{240-k} P(N(m) = l) P(N(m+3) - N(m) = k) P(N(600) - N(m+3) = 240 - k - l) \\ &= \sum_{l=0}^{240-k} \frac{(m\lambda)^l}{l!} e^{-m\lambda} \cdot \frac{(3\lambda)^k}{k!} e^{-3\lambda} \cdot \frac{[\lambda(597-m)]^{240-k-l}}{(240-k-l)!} e^{-\lambda(597-m)} \\ &= \lambda^{240} e^{-600\lambda} \frac{3^k}{k!} \sum_{l=0}^{240-k} \frac{m^l (597-m)^{240-k-l}}{l! (240-k-l)!} \\ &= \lambda^{240} e^{-600\lambda} \frac{3^k 597^{240-k}}{k! (240-k)!}, \quad k = 0, 1, 2, \dots, 240, m = 0, 1, 2, \dots, 597, \end{aligned}$$

$$P(N(600) = 240) = \frac{(600\lambda)^{240}}{240!} e^{-600\lambda},$$

因此, 由 (1) 可得

$$\begin{aligned} & P(N(m+3) - N(m) = k \mid N(600) = 240) \\ &= \frac{P(N(m+3) - N(m) = k, N(600) = 240)}{P(N(600) = 240)} \\ &= \frac{240!}{k! (240-k)!} \left(\frac{3}{600}\right)^k \left(1 - \frac{3}{600}\right)^{240-k}, \\ & \quad k = 0, 1, 2, \dots, 240, m = 0, 1, 2, \dots, 597. \end{aligned} \quad (2)$$

利用以下事实:

$$\lim_{n \rightarrow \infty, np_n \rightarrow \lambda} \frac{n!}{k! (n-k)!} p_n^k (1-p_n)^{n-k} = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \dots \quad (3)$$

可将 (2) 中概率表为

$$\begin{aligned} & P(N(m+3) - N(m) = k \mid N(600) = 240) \\ & \approx \frac{(240 \cdot 3/600)^k}{k!} e^{-240 \cdot 3/600} = \frac{(6/5)^k}{k!} e^{-6/5}, \end{aligned}$$

$$k = 0, 1, 2, \dots, 240, m = 0, 1, 2, \dots, 597. \quad (4)$$

根据 (2) 和 (4) 计算可得

k	精确值	近似值	k	精确值	近似值
0	0.3003	0.3012	6	0.0012	0.0012
1	0.3622	0.3614	7	0.0002	0.0002
2	0.2175	0.2169	8	0.0000	0.0000
3	0.0867	0.0867	9	0.0000	0.0000
4	0.0258	0.0260	10	0.0000	0.0000
5	0.0061	0.0062	11	0.0000	0.0000

■

8. 令 $\{N_i(t) : t \geq 0\}, i = 1, \dots, n$ 为 n 个独立的分别有强度参数 $\lambda_i, i = 1, \dots, n$ 的 Poisson 过程, 记 T 为在全部 n 个过程中至少发生了一件事实的时刻, 试求 T 的分布.

解 由题意可知

$$\{T > t\} = \{N_i(t) = 0, i = 1, \dots, n\}, \quad t > 0,$$

故

$$\begin{aligned} P(T > t) &= P(N_i(t) = 0, i = 1, \dots, n) \\ &= P(N_1(t) = 0) \cdots P(N_n(t) = 0) \\ &= e^{-t \sum_{i=1}^n \lambda_i}, \quad t > 0, \end{aligned}$$

因而, T 的分布函数为

$$\begin{aligned} F_T(t) &= 1 - P(T > t) \\ &= 1 - e^{-t \sum_{i=1}^n \lambda_i}, \quad t > 0, \end{aligned}$$

密度函数为

$$f_T(t) = \frac{dF_T(t)}{dt} = \sum_{i=1}^n \lambda_i \cdot e^{-t \sum_{i=1}^n \lambda_i}, \quad t > 0.$$

这说明了 T 服从参数为 $\sum_{i=1}^n \lambda_i$ 的指数分布. ■

8. 设 $\{N(t) : t \geq 0\}$ 是强度为 λ 的 Poisson 过程. 给定 $N(t) = n$, 试求第 r 个事件发生的时刻 W_r 的条件密度函数 $f_{W_r|N(t)}(w_r | n), r = 1, \dots, n$.

解 注意到

$$\{W_r \leq w_r\} = \{N(w_r) \geq r\}, \quad r = 1, \dots, n,$$

因而有

$$P(W_r \leq w_r | N(t) = n) = P(N(w_r) \geq r | N(t) = n)$$

$$= \frac{P(N(w_r) \geq r, N(t) = n)}{P(N(t) = n)}, \quad w_r, t > 0. \quad (1)$$

而当 $0 < w_r < t$ 时, 有

$$\begin{aligned} & P(N(w_r) \geq r, N(t) = n) \\ &= \sum_{k=r}^n P(N(w_r) = k, N(t) = n) \\ &= \sum_{k=r}^n P(N(w_r) = k, N(t) - N(w_r) = n - k) \\ &= \sum_{k=r}^n P(N(w_r) = k)P(N(t) - N(w_r) = n - k) \\ &= \sum_{k=r}^n \frac{(\lambda w_r)^k}{k!} e^{-\lambda w_r} \cdot \frac{[\lambda(t - w_r)]^{n-k}}{(n-k)!} e^{-\lambda(t-w_r)} \\ &= \lambda^n e^{-\lambda t} \sum_{k=r}^n \frac{w_r^k (t - w_r)^{n-k}}{k!(n-k)!}, \quad r = 1, \dots, n. \end{aligned} \quad (2)$$

又

$$P(N(t) = n) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, \quad n = 0, 1, 2, \dots. \quad (3)$$

将 (2) 和 (3) 代入 (1) 中可得

$$\begin{aligned} P(W_r \leq w_r \mid N(t) = n) &= \frac{1}{t^n} \sum_{k=r}^n \frac{n!}{k!(n-k)!} w_r^k (t - w_r)^{n-k}, \\ & \quad 0 < w_r < t, r = 1, \dots, n. \end{aligned} \quad (4)$$

而

$$\begin{aligned} \int_0^{w_r} u^{r-1} (t - u)^{n-r} du &= \frac{1}{r} \int_0^{w_r} (t - u)^{n-r} d(u^r) \\ &= \frac{1}{r} w_r^r (t - w_r)^{n-r} + \frac{n-r}{r} \int_0^{w_r} u^r (t - u)^{n-r-1} du \\ &= \dots \\ &= (r-1)!(n-r)! \sum_{k=r}^n \frac{w_r^r (t - w_r)^{n-r}}{k!(n-k)!}, \\ & \quad 0 < w_r < t, r = 1, \dots, n, \end{aligned}$$

将其代入 (4) 中得

$$\begin{aligned} P(W_r \leq w_r \mid N(t) = n) &= \frac{n!}{(r-1)!(n-r)!t^n} \int_0^{w_r} u^{r-1} (t - u)^{n-r} du, \\ & \quad 0 < w_r < t, r = 1, \dots, n. \end{aligned}$$

这说明了, 定 $N(t) = n, W_r$ 的条件密度函数为

$$f_{W_r|N(t)}(w_r \mid n) = \frac{n!}{(r-1)!(n-r)!t^n} w_r^{r-1} (t - w_r)^{n-r}, \quad 0 < w_r < t, r = 1, \dots, n.$$

9. 考虑参数为 λ 的 Poisson 过程 $\{N(t) : t \geq 0\}$, 若每一事件独立地以概率 p 被观察到, 并将观察到的过程记为 $\{N_1(t) : t \geq 0\}$. 试问 $\{N_1(t) : t \geq 0\}$ 是什么过程? $\{N(t) - N_1(t) : t \geq 0\}$ 呢? $\{N_1(t) : t \geq 0\}$ 与 $\{N(t) - N_1(t) : t \geq 0\}$ 是否独立?

解 由题设易知

$$(i) N_1(0) = 0;$$

(ii) $\{N_1(t) : t \geq 0\}$ 是一独立增量过程.

往证

(iii) 对 $0 \leq s < t$, $N_1(t) - N_1(s)$ 服从参数为 $\lambda p(t-s)$ 的 Poisson 分布. 从而, 由 (i)–(iii) 可知, $\{N_1(t) : t \geq 0\}$ 是一参数为 λp 的 Poisson 过程. 其实, 对 $0 \leq s < t$, 有

$$P(N_1(t) - N_1(s) = m \mid N(t) - N(s) = n) = \binom{n}{m} p^m (1-p)^{n-m},$$

$$m = 0, 1, \dots, n, n = 0, 1, 2, \dots \quad (1)$$

而由 $\{N(t) : t \geq 0\}$ 是参数为 λ 的 Poisson 过程可知, 对 $0 \leq s < t$, 有

$$P(N(t) - N(s) = n) = \frac{[\lambda(t-s)]^n}{n!} e^{-\lambda(t-s)}, \quad n = 0, 1, 2, \dots \quad (2)$$

故由 (1) 和 (2) 可得

$$\begin{aligned} & P(N_1(t) - N_1(s) = m) \\ &= \sum_{n=m}^{\infty} P(N_1(t) - N_1(s) = m \mid N(t) - N(s) = n) P(N(t) - N(s) = n) \\ &= \sum_{n=m}^{\infty} \binom{n}{m} p^m (1-p)^{n-m} \cdot \frac{[\lambda(t-s)]^n}{n!} e^{-\lambda(t-s)} \\ &= \frac{p^m}{m!} e^{-\lambda(t-s)} \sum_{n=m}^{\infty} \frac{(1-p)^{n-m} [\lambda(t-s)]^n}{(n-m)!} \\ &= \frac{[\lambda p(t-s)]^m}{m!} e^{-\lambda p(t-s)}, \quad m = 0, 1, \dots, \end{aligned}$$

这即说明了, 对 $0 \leq s < t$, $N_1(t) - N_1(s)$ 服从参数为 $\lambda p(t-s)$ 的 Poisson 分布.

注意到若 n 个随机事件 $A_1, \dots, A_n$ 独立, 则 n 个随机事件 $\bar{A}_1, \dots, \bar{A}_n$ 也独立, 其中 $\bar{A}_i$ 等于 A_i 或等于 A_i 的对立事件, $i = 1, \dots, n$, 因而, $\{N_1(t) : t \geq 0\}$ 与 $\{N(t) - N_1(t) : t \geq 0\}$ 独立. 由 $N_1(t)$ 和 $N(t) - N_1(t)$ 的对称性可知, $\{N(t) - N_1(t) : t \geq 0\}$ 是一参数为 $\lambda(1-p)$ 的 Poisson 过程. ■

10. 公路上到达某加油站的卡车服从参数为 λ_1 的 Poisson 过程 $\{N_1(t) : t \geq 0\}$, 而到达的小汽车服从参数为 λ_2 的 Poisson 过程 $\{N_2(t) : t \geq 0\}$, 且 $\{N_1(t) : t \geq 0\}$ 与 $\{N_2(t) : t \geq 0\}$ 独立. 试问 $\{N(t) : t \geq 0\} \triangleq \{N_1(t) + N_2(t) : t \geq 0\}$ 是什么过程? 并计算在总车流数 $\{N(t) : t \geq 0\}$ 中, 卡车首先到达的概率.

解 首先, 由题设易知, $N(0) = 0$.

其次, 由于 $\{N_1(t) : t \geq 0\}$ 与 $\{N_2(t) : t \geq 0\}$ 独立, 因此, 对任意 $0 \leq t_1 < \cdots < t_n, n$ 维随机向量 $(N_1(t_1), \cdots, N_1(t_n))$ 与 $(N_2(t_1), \cdots, N_2(t_n))$ 独立, 从而, $n-1$ 维随机向量 $(N_1(t_2) - N_1(t_1), \cdots, N_1(t_n) - N_1(t_{n-1}))$ 与 $(N_2(t_2) - N_2(t_1), \cdots, N_2(t_n) - N_2(t_{n-1}))$ 独立, 而 $N_1(t_2) - N_1(t_1), \cdots, N_1(t_n) - N_1(t_{n-1})$ 独立, $N_2(t_2) - N_2(t_1), \cdots, N_2(t_n) - N_2(t_{n-1})$ 独立, 故 $N_1(t_2) - N_1(t_1), \cdots, N_1(t_n) - N_1(t_{n-1}), N_2(t_2) - N_2(t_1), \cdots, N_2(t_n) - N_2(t_{n-1})$ 独立, 因而, $N(t_2) - N(t_1), \cdots, N(t_n) - N(t_{n-1})$ 独立. 这说明了, $\{N(t) : t \geq 0\}$ 是一独立增量过程.

最后, 对任意 $0 \leq s < t$, 由于 $N_1(t) - N_1(s)$ 服从参数为 $\lambda_1(t-s)$ 的 Poisson 分布, $N_2(t) - N_2(s)$ 服从参数为 $\lambda_2(t-s)$ 的 Poisson 分布, 且 $N_1(t) - N_1(s)$ 与 $N_2(t) - N_2(s)$ 独立, 因此, $N(t) - N(s)$ 服从参数为 $(\lambda_1 + \lambda_2)(t-s)$ 的 Poisson 分布. 故 $\{N(t) : t \geq 0\}$ 是一参数为 $\lambda_1 + \lambda_2$ 的 Poisson 过程.

以 U_1 和 V_1 分别表示在过程 $\{N_1(t) : t \geq 0\}$ 和 $\{N_2(t) : t \geq 0\}$ 中第一辆车达到的时刻, 则 U_1 和 V_1 独立, 分别服从参数为 λ_1 和 λ_2 的指数分布. 所求卡车首先到达的概率为

$$\begin{aligned} P(U_1 < V_1) &= \lambda_1 \lambda_2 \iint_{u, v > 0, u < v} e^{-(\lambda_1 u + \lambda_2 v)} du dv \\ &= \lambda_1 \lambda_2 \int_0^{+\infty} e^{-\lambda_1 u} du \int_u^{+\infty} e^{-\lambda_2 v} dv \\ &= \lambda_1 \int_0^{+\infty} e^{-(\lambda_1 + \lambda_2)u} du \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2}. \end{aligned}$$

■

11. 冲击模型 (Shock Model) 记 $N(t)$ 为某系统到时刻 t 受到的冲击次数, 这是参数为 λ 的 Poisson 过程. 设第 k 次冲击对系统的损害大小 Y_k 服从参数为 μ 的指数分布, $Y_k, k = 1, 2, \cdots$ 独立同分布. 记 $X(t)$ 为系统所受到的总损害, 当损害超过一定的极限 α 时, 系统就不能运行, 寿命终止. 记 T 为系统寿命. 试求该系统的平均寿命 $E(T)$, 并对所得结果作出直观解释.

解 注意到

$$\{T > t\} = \{X(t) \leq \alpha\}, \quad t > 0,$$

因而

$$\begin{aligned} P(T > t) &= P(X(t) \leq \alpha) = P\left(\sum_{k=1}^{N(t)} Y_k \leq \alpha\right) \\ &= \sum_{n=1}^{\infty} P\left(\sum_{k=1}^{N(t)} Y_k \leq \alpha, N(t) = n\right) \\ &= \sum_{n=1}^{\infty} P\left(\sum_{k=1}^{N(t)} Y_k \leq \alpha \mid N(t) = n\right) P(N(t) = n) \\ &= \sum_{n=1}^{\infty} P\left(\sum_{k=1}^n Y_k \leq \alpha \mid N(t) = n\right) P(N(t) = n) \end{aligned}$$

$$= \sum_{n=1}^{\infty} P\left(\sum_{k=1}^n Y_k \leq \alpha\right) P(N(t) = n), \quad t > 0. \quad (1)$$

而由于 $Y_k, k = 1, 2, \dots$ 独立同分布, Y_1 服从参数为 μ 的指数分布, 因此, $S_n \triangleq \sum_{k=1}^n Y_k$ 具有密度函数

$$f_{S_n}(s) = \frac{\mu^n}{(n-1)!} s^{n-1} e^{-\mu s}, \quad s > 0, n = 1, 2, \dots,$$

故

$$P\left(\sum_{k=1}^n Y_k \leq \alpha\right) = \frac{\mu^n}{(n-1)!} \int_0^\alpha s^{n-1} e^{-\mu s} ds, \quad n = 1, 2, \dots. \quad (2)$$

又

$$P(N(t) = n) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, \quad n = 0, 1, 2, \dots. \quad (3)$$

将 (2) 和 (3) 代入 (1) 中得

$$P(T > t) = e^{-\lambda t} \sum_{n=1}^{\infty} \frac{(\lambda \mu t)^n}{n!(n-1)!} \int_0^\alpha s^{n-1} e^{-\mu s} ds, \quad t > 0.$$

故

$$\begin{aligned} E(T) &= \int_0^{+\infty} P(T > t) dt \\ &= \sum_{n=1}^{\infty} \frac{(\mu \lambda)^n}{n!(n-1)!} \int_0^{+\infty} t^n e^{-\lambda t} dt \cdot \int_0^\alpha s^{n-1} e^{-\mu s} ds \\ &= \frac{1}{\lambda} \sum_{n=1}^{\infty} \frac{\mu^n}{(n-1)!} \int_0^\alpha s^{n-1} e^{-\mu s} ds \\ &= \frac{1}{\lambda} \int_0^\alpha e^{-\mu s} \sum_{n=1}^{\infty} \frac{\mu^n}{(n-1)!} s^{n-1} ds \\ &= \frac{\mu}{\lambda} \int_0^\alpha ds = \frac{\mu \alpha}{\lambda}. \end{aligned}$$

这说明了, 若 λ 越大 (即系统所受冲击越频繁), μ 越小 (即每次冲击所造成的平均损害越大), α 越小 (即系统所能承受的损害极限越小), 则系统的平均寿命越短. 这是符合常识的. ■

12. 令 $\{N(t) : t \geq 0\}$ 是强度函数为 $\lambda(t)$ 的非齐次 Poisson 过程, $X_1, X_2, \dots$ 为事件间的时间间隔.

- (i) $X_1, X_2, \dots$ 是否独立?
- (ii) $X_1, X_2, \dots$ 是否同分布?
- (iii) 试求 (X_1, X_2) 的分布.

解 注意到

$$\{W_k \leq t\} = \{N(t) \geq k\}, \quad k = 1, 2, \dots,$$

因此, (W_1, W_2) 的联合分布函数为

$$\begin{aligned}
 F_{(W_1, W_2)}(t_1, t_2) &= P(W_1 \leq t_1, W_2 \leq t_2) \\
 &= P(N(t_1) \geq 1, N(t_2) \geq 2) \\
 &= \sum_{k=2}^{\infty} \sum_{l=1}^k P(N(t_1) = l, N(t_2) = k), \quad 0 \leq t_1 < t_2.
 \end{aligned} \tag{1}$$

而对 $1 \leq l \leq k, 0 \leq t_1 < t_2$, 有

$$\begin{aligned}
 P(N(t_1) = l, N(t_2) = k) &= P(N(t_1) = l, N(t_2) - N(t_1) = k - l) \\
 &= P(N(t_1) = l)P(N(t_2) - N(t_1) = k - l) \\
 &= \frac{m(t_1)^l}{l!} e^{-m(t_1)} \cdot \frac{[m(t_2) - m(t_1)]^{k-l}}{(k-l)!} e^{-[m(t_2) - m(t_1)]} \\
 &= \frac{m(t_1)^l [m(t_2) - m(t_1)]^{k-l}}{l!(k-l)!} e^{-m(t_2)},
 \end{aligned}$$

将其代入 (1) 中得

$$\begin{aligned}
 F_{(W_1, W_2)}(t_1, t_2) &= e^{-m(t_2)} \sum_{k=2}^{\infty} \frac{1}{k!} \sum_{l=1}^k \binom{k}{l} m(t_1)^l [m(t_2) - m(t_1)]^{k-l} \\
 &= e^{-m(t_2)} \sum_{k=2}^{\infty} \frac{1}{k!} \left\{ \sum_{l=0}^k \binom{k}{l} m(t_1)^l [m(t_2) - m(t_1)]^{k-l} - [m(t_2) - m(t_1)]^k \right\} \\
 &= e^{-m(t_2)} \sum_{k=2}^{\infty} \frac{1}{k!} \{ m(t_2)^k - [m(t_2) - m(t_1)]^k \} \\
 &= e^{-m(t_2)} \left\{ \sum_{k=0}^{\infty} \frac{m(t_2)^k - [m(t_2) - m(t_1)]^k}{k!} - m(t_1) \right\} \\
 &= e^{-m(t_2)} \{ e^{m(t_2)} - e^{m(t_2) - m(t_1)} - m(t_1) \} \\
 &= 1 - e^{-m(t_1)} - m(t_1)e^{-m(t_2)}, \quad 0 \leq t_1 < t_2.
 \end{aligned} \tag{2}$$

故 (W_1, W_2) 的联合密度函数为

$$\begin{aligned}
 f_{(W_1, W_2)}(t_1, t_2) &= \frac{\partial^2 F_{(W_1, W_2)}(t_1, t_2)}{\partial t_1 \partial t_2} \\
 &= \lambda(t_1)\lambda(t_2)e^{-m(t_2)}, \quad 0 \leq t_1 < t_2.
 \end{aligned} \tag{3}$$

由于

$$W_1 = X_1, \quad W_2 = X_1 + X_2,$$

因此从 (W_1, W_2) 到 (X_1, X_2) 的 Jacob 行列式为

$$\frac{\partial(W_1, W_2)}{\partial(X_1, X_2)} = 1,$$

因而, 由 (3) 可得 (X_1, X_2) 的联合密度函数为

$$f_{(X_1, X_2)}(t_1, t_2) = \lambda(t_1)\lambda(t_1 + t_2)e^{-m(t_1 + t_2)}, \quad t_1, t_2 > 0. \tag{4}$$

这说明了,一般地, X_1 与 X_2 不独立, 且 X_1 的密度函数为

$$\begin{aligned} f_{X_1}(t_1) &= \lambda(t_1) \int_0^{+\infty} \lambda(t_1 + t_2) e^{-m(t_1+t_2)} dt_2 \\ &= \lambda(t_1) [e^{-m(t_1)} - e^{-m(+\infty)}], \quad t_1 > 0, \end{aligned} \quad (5)$$

其中 $e^{-m(+\infty)}$ 由下式确定:

$$\begin{aligned} 1 &= \int_0^{+\infty} f_{X_1}(t_1) dt_1 = \int_0^{+\infty} \lambda(t_1) [e^{-m(t_1)} - e^{-m(+\infty)}] dt_1 \\ &= 1 - [m(+\infty) + 1] e^{-m(+\infty)}, \end{aligned}$$

即

$$e^{-m(+\infty)} = 0.$$

将其代入 (5) 中得

$$f_{X_1}(t_1) = \lambda(t_1) e^{-m(t_1)}, \quad t_1 > 0, \quad (6)$$

进而, 由 (4) 知, X_2 的密度函数为

$$f_{X_2}(t_2) = \int_0^{+\infty} \lambda(t_1) \lambda(t_1 + t_2) e^{-m(t_1+t_2)} dt_1, \quad t_2 > 0, \quad (7)$$

■

13. 考虑对所有 $t \geq 0$, 强度函数 $\lambda(t)$ 均大于 0 的非齐次 Poisson 过程 $\{N(t), t \geq 0\}$. 令 $m(t) = \int_0^t \lambda(u) du, t \geq 0$, $m(t)$ 的反函数记为 $l(t), t \geq 0$, 记 $N_1(t) = N(l(t)), t \geq 0$. 试证: $\{N_1(t), t \geq 0\}$ 是通常的 Poisson 过程, 并求其强度.

解 首先, $N_1(0) = N(l(0)) = N(0) = 0$. 其次, 由题设可知, $m(t)$ 是 $t \geq 0$ 的严增函数, 因此其反函数 $l(t)$ 也是 $t \geq 0$ 的严增函数. 从而, 对任意 $0 \leq t_1 < t_2 < \cdots < t_n$, 有 $0 \leq l(t_1) < l(t_2) < \cdots < l(t_n)$, 故由

$$N_1(t_2) - N_1(t_1) = N(l(t_2)) - N(l(t_1)), \cdots, N_1(t_n) - N_1(t_{n-1}) = N(l(t_n)) - N(l(t_{n-1}))$$

及 $\{N(t), t \geq 0\}$ 中增量的独立性可知, $\{N_1(t), t \geq 0\}$ 中增量也是独立的. 最后, 对任意 $0 \leq s < t$, 有

$$\begin{aligned} P(N_1(t) - N_1(s) = k) &= P(N(l(t)) - N(l(s)) = k) \\ &= \frac{[m(l(t)) - m(l(s))]^k}{k!} e^{-[m(l(t)) - m(l(s))]}, \quad k = 0, 1, 2, \cdots \end{aligned}$$

而

$$m(l(t)) = t, \quad t \geq 0,$$

故

$$P(N_1(t) - N_1(s) = k) = \frac{(t-s)^k}{k!} e^{-(t-s)}, \quad k = 0, 1, 2, \cdots$$

综上所述, $\{N(t), t \geq 0\}$ 是一强度为 1 的 Poisson 过程. ■

14. 设 $\{N(t), t \geq 0\}$ 是一更新过程, 试判断下述命题的真伪:

$$(i) \{N(t) < k\} = \{W_k > t\},$$

$$(ii) \{N(t) \leq k\} = \{W_k \geq t\},$$

$$(iii) \{N(t) > k\} = \{W_k < t\},$$

其中 W_k 是第 k 个事件的等待时间, $k = 1, 2, \dots$.

解 由 $\{W_k \leq t\} = \{N(t) \geq k\}, k = 1, 2, \dots, t \geq 0$ 可知

$$(i) \{N(t) < k\} = \overline{\{N(t) \geq k\}} = \overline{\{W_k \leq t\}} = \{W_k > t\}.$$

$$(ii) \{N(t) \leq k\} = \{N(t) < k+1\} = \overline{\{N(t) \geq k+1\}} = \overline{\{W_{k+1} \leq t\}} = \{W_{k+1} > t\}.$$

$$(iii) \{N(t) > k\} = \{N(t) \geq k+1\} = \{W_{k+1} \leq t\}. \blacksquare$$

习题 3

1. 对于 Markov 链 $\{X_n, n = 0, 1, 2, \dots\}$, 试证条件

$$P(X_{n+1} = j \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i) = P(X_{n+1} = j \mid X_n = i)$$

等价于对所有非负整数 n 和 m 及所有状态 $i_0, \dots, i_n, j_1, \dots, j_m$, 有

$$\begin{aligned} & P(X_{n+1} = j_1, \dots, X_{n+m} = j_m \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) \\ &= P(X_{n+1} = j_1, \dots, X_{n+m} = j_m \mid X_n = i_n). \end{aligned} \quad (0)$$

证明 首先证明 $\{X_n, n = 0, 1, 2, \dots\}$ 为 Markov 链 $\Leftrightarrow$ 对任意非负整数 n 和 $k \leq n$, 任意 $\{n_1, \dots, n_k\} \subset \{0, 1, \dots, n-1\}$ 及任意状态 $i_{n_1}, \dots, i_{n_k}, i, j$, 均有

$$\begin{aligned} & P(X_{n+1} = j \mid X_{n_1} = i_{n_1}, \dots, X_{n_k} = i_{n_k}, X_n = i) \\ &= P(X_{n+1} = j \mid X_n = i). \end{aligned} \quad (1)$$

其实,“(1) $\Rightarrow$ { $X_n, n = 0, 1, 2, \dots$ } 为 Markov 链”是显然的(在(1)中取 $k = n$ 即可看出). 往证”{ $X_n, n = 0, 1, 2, \dots$ } 为 Markov 链 $\Rightarrow$ (1)”. 由 $\{X_n, n = 0, 1, 2, \dots\}$ 为 Markov 链知, 对任意非负整数 n 及任意状态 $i_0, i_1, \dots, i_{n-1}, i, j$, 均有

$$P(X_{n+1} = j \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i) = P(X_{n+1} = j \mid X_n = i),$$

即

$$\frac{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i, X_{n+1} = j)}{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i)} = \frac{P(X_n = i, X_{n+1} = j)}{P(X_n = i)},$$

因而

$$\frac{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i, X_{n+1} = j)}{P(X_n = i, X_{n+1} = j)} = \frac{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i)}{P(X_n = i)}. \quad (2)$$

而

$$\begin{aligned} & P(X_{n_1} = i_{n_1}, \dots, X_{n_k} = i_{n_k}, X_n = i, X_{n+1} = j) \\ &= \sum_{\substack{i_m \in \mathcal{X}, m = 0, 1, \dots, n-1, \\ m \neq n_1, \dots, n_k}} P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i, X_{n+1} = j), \\ & P(X_{n_1} = i_{n_1}, \dots, X_{n_k} = i_{n_k}, X_n = i) \\ &= \sum_{\substack{i_m \in \mathcal{X}, m = 0, 1, \dots, n-1, \\ m \neq n_1, \dots, n_k}} P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i) \end{aligned}$$

因而由(2)可得

$$\frac{P(X_{n_1} = i_{n_1}, \dots, X_{n_k} = i_{n_k}, X_n = i, X_{n+1} = j)}{P(X_n = i, X_{n+1} = j)} = \frac{P(X_{n_1} = i_{n_1}, \dots, X_{n_k} = i_{n_k}, X_n = i)}{P(X_n = i)},$$

即

$$\frac{P(X_{n_1} = i_{n_1}, \dots, X_{n_k} = i_{n_k}, X_n = i, X_{n+1} = j)}{P(X_{n_1} = i_{n_1}, \dots, X_{n_k} = i_{n_k}, X_n = i)} = \frac{P(X_n = i, X_{n+1} = j)}{P(X_n = i)},$$

由此立得 (1).

其次, 证明 $\{X_n, n = 0, 1, 2, \dots\}$ 为 Markov 链 $\Leftrightarrow (0)$. ”(0) $\Rightarrow$ ” $\{X_n, n = 0, 1, 2, \dots\}$ 为 Markov 链”是显然的 (在 (0) 中取 $m = 1$ 即可看出). 往证” $\{X_n, n = 0, 1, 2, \dots\}$ 为 Markov 链 $\Rightarrow (0)$ ”. 由 $\{X_n, n = 0, 1, 2, \dots\}$ 为 Markov 链可得

$$\begin{aligned} & P(X_{n+1} = j_1, \dots, X_{n+m} = j_m \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) \\ &= \frac{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n, X_{n+1} = j_1, \dots, X_{n+m} = j_m)}{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n)} \\ &= \frac{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n, X_{n+1} = j_1, \dots, X_{n+m} = j_m)}{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n, X_{n+1} = j_1, \dots, X_{n+m-1} = j_{m-1})} \\ & \quad \cdot \frac{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n, X_{n+1} = j_1, \dots, X_{n+m-1} = j_{m-1})}{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n)} \\ &= P(X_{n+m} = j_m \mid X_{n+m-1} = j_{m-1}) \\ & \quad \cdot \frac{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n, X_{n+1} = j_1, \dots, X_{n+m-1} = j_{m-1})}{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n)} \\ &= \dots \\ &= P(X_{n+m} = j_m \mid X_{n+m-1} = j_{m-1}) \cdots P(X_{n+2} = j_2 \mid X_{n+1} = j_1) P(X_{n+1} = j_1 \mid X_n = i_n). \end{aligned} \quad (3)$$

由已证 (1) 可得

$$\begin{aligned} & P(X_{n+1} = j_1, \dots, X_{n+m} = j_m \mid X_n = i_n) \\ &= \frac{P(X_n = i_n, X_{n+1} = j_1, \dots, X_{n+m} = j_m)}{P(X_n = i_n)} \\ &= \frac{P(X_n = i_n, X_{n+1} = j_1, \dots, X_{n+m} = j_m)}{P(X_n = i_n, X_{n+1} = j_1, \dots, X_{n+m-1} = j_{m-1})} \\ & \quad \cdot \frac{P(X_n = i_n, X_{n+1} = j_1, \dots, X_{n+m-1} = j_{m-1})}{P(X_n = i_n)} \\ &= P(X_{n+m} = j_m \mid X_{n+m-1} = j_{m-1}) \\ & \quad \cdot \frac{P(X_n = i_n, X_{n+1} = j_1, \dots, X_{n+m-1} = j_{m-1})}{P(X_n = i_n)} \\ &= \dots \\ &= P(X_{n+m} = j_m \mid X_{n+m-1} = j_{m-1}) \cdots P(X_{n+2} = j_2 \mid X_{n+1} = j_1) P(X_{n+1} = j_1 \mid X_n = i_n). \end{aligned} \quad (4)$$

由 (3) 和 (4) 即得 (0). ■

2. 考虑状态空间 $\mathcal{X} = \{0, 1, 2\}$ 上的一个 Markov 链 $\{X_n, n = 0, 1, 2, \dots\}$, 其转移概

率矩阵为

$$P = \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.9 & 0.1 & 0 \\ 0.1 & 0.8 & 0.1 \end{pmatrix},$$

初始分布为 $p_0 = 0.3, p_1 = 0.4, p_2 = 0.3$, 试求概率 $P(X_0 = 0, X_1 = 1, X_2 = 2)$.

解 所求概率为

$$\begin{aligned} & P(X_0 = 0, X_1 = 1, X_2 = 2) \\ &= P(X_2 = 2 \mid X_1 = 1)P(X_1 = 1 \mid X_0 = 0)P(X_0 = 0) \\ &= P_{12}P_{01}p_0 = 0. \end{aligned}$$

■

3. 信号传送问题. 信号只有 0, 1 两种, 分为多个阶段传送. 在每一步上出错的概率为 α . $X_0 = 0$ 是送出的信号, 而 X_n 是在第 n 步接收到的信号. 假定 $\{X_n, n = 0, 1, 2, \dots\}$ 为一 Markov 链, 其状态空间为 $\mathcal{X} = \{0, 1\}$, 转移概率矩阵为

$$P = \begin{pmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{pmatrix} = \begin{pmatrix} 1 - \alpha & \alpha \\ \alpha & 1 - \alpha \end{pmatrix}, \quad \alpha \in (0, 1).$$

试求

(a) n 步均不出错的概率 $P(X_1 = 0, \dots, X_n = 0 \mid X_0 = 0), n = 1, 2, \dots$,

(b) n 步之后传送无误的概率 $P(X_n = 0 \mid X_0 = 0), n = 1, 2, \dots$.

解 (a) 所求概率为

$$\begin{aligned} & P(X_1 = 0, \dots, X_n = 0 \mid X_0 = 0) \\ &= P(X_n = 0 \mid X_0 = 0, X_1 = 0, \dots, X_{n-1} = 0)P(X_1 = 0, \dots, X_{n-1} = 0 \mid X_0 = 0) \\ &= P(X_n = 0 \mid X_{n-1} = 0)P(X_1 = 0, \dots, X_{n-1} = 0 \mid X_0 = 0) \\ &= \dots\dots\dots \\ &= P(X_n = 0 \mid X_{n-1} = 0) \cdots P(X_1 = 0 \mid X_0 = 0) \\ &= (1 - \alpha)^n, \quad n = 1, 2, \dots \end{aligned}$$

(b) 所求概率为

$$P(X_n = 0 \mid X_0 = 0) = P_{00}^{(n)}, \quad n = 1, 2, \dots, \quad (1)$$

其中 $P_{ij}^{(n)}$ 为 n 步转移概率矩阵

$$P^{(n)} = \begin{pmatrix} P_{00}^{(n)} & P_{01}^{(n)} \\ P_{10}^{(n)} & P_{11}^{(n)} \end{pmatrix} = P^n = \begin{pmatrix} 1 - \alpha & \alpha \\ \alpha & 1 - \alpha \end{pmatrix}^n \quad (2)$$

的元素, $i, j = 0, 1$. 注意到

$$P = T \begin{pmatrix} 1 & 0 \\ 0 & 1 - 2\alpha \end{pmatrix} T',$$

其中

$$T = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

为一正交阵, 因而由 (2) 得

$$\begin{aligned} P^{(n)} &= T \begin{pmatrix} 1 & 0 \\ 0 & 1-2\alpha \end{pmatrix}^n T' = T \begin{pmatrix} 1 & 0 \\ 0 & (1-2\alpha)^n \end{pmatrix} T' \\ &= \frac{1}{2} \begin{pmatrix} 1+(1-2\alpha)^n & 1-(1-2\alpha)^n \\ 1-(1-2\alpha)^n & 1+(1-2\alpha)^n \end{pmatrix}, \quad n=1, 2, \dots \end{aligned} \quad (3)$$

由此及 (1) 即得

$$P(X_n = 0 \mid X_0 = 0) = \frac{1}{2}[1 + (1-2\alpha)^n], \quad n=1, 2, \dots$$

■

4. A, B 两罐总共装着 N 个球. 作如下实验: 在时刻 n 先从 N 个球中等概率地任取一球, 然后从 A, B 两罐中任选一罐, 选中 A 罐的概率为 p , 选中 B 罐的概率为 q , $p+q=1$, 之后再将选出的球放入选好的罐中. 设 X_n 为时刻 n 时 A 罐中的球数, 试求此 Markov 链 $\{X_n, n=0, 1, 2, \dots\}$ 的转移概率矩阵.

解证 由题设知, $\{X_n, n=0, 1, 2, \dots\}$ 的状态空间为 $\mathcal{X} = \{0, 1, 2, \dots, N\}$. 以 I_n 表示在时刻 n 从 N 个球中等概率地取得一球的结果, 约定

$$I_n = \begin{cases} 0, & \text{在时刻 } n \text{ 从 B 罐中取出一球,} \\ -1, & \text{在时刻 } n \text{ 从 A 罐中取出一球.} \end{cases}$$

由题设可知, 给定 $X_k = i_k, k \in \mathcal{X}, k=0, 1, \dots, n-1$ 时, I_n 的条件分布为

$$\begin{cases} P(I_n = 0 \mid X_k = i_k, k=0, 1, \dots, n-1) = P(I_n = 0 \mid X_{n-1} = i_{n-1}) = 1 - \frac{i_{n-1}}{N}, \\ P(I_n = -1 \mid X_k = i_k, k=0, 1, \dots, n-1) = P(I_n = -1 \mid X_{n-1} = i_{n-1}) = \frac{i_{n-1}}{N}. \end{cases} \quad (1)$$

再以 J_n 表示在时刻 n 从 A, B 两罐中任选一罐所得的结果, 约定

$$J_n = \begin{cases} 1, & \text{在时刻 } n \text{ 选中 A 罐,} \\ 0, & \text{在时刻 } n \text{ 选中 B 罐.} \end{cases}$$

由题设可知, $\{X_n, n=0, 1, 2, \dots, I_n = 1, 2, \dots\}$ 与 $\{J_n, n=1, 2, \dots\}$ 独立, 且 J_n 的分布为

$$\begin{cases} P(J_n = 1) = p, \\ P(J_n = 0) = q. \end{cases} \quad (2)$$

此外, 有

$$X_n = X_{n-1} + I_n + J_n, \quad n=1, 2, \dots \quad (3)$$

下面往证 $\{X_n, n = 0, 1, 2, \dots\}$ 为一 Markov 链. 其实, 由 (1) 和 (2) 可知, 对任意正整数 n 及任意状态 $i_0, \dots, i_{n+1} \in \mathcal{X}$, 有

$$\begin{aligned}
 & P(X_{n+1} = i_{n+1} \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) \\
 &= P(X_n + I_{n+1} + J_{n+1} = i_{n+1} \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) \\
 &= P(I_{n+1} + J_{n+1} = i_{n+1} - i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) \\
 &= P(I_{n+1} + J_{n+1} = i_{n+1} - i_n, I_{n+1} = 0 \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) \\
 &\quad + P(I_{n+1} + J_{n+1} = i_{n+1} - i_n, I_{n+1} = -1 \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) \\
 &= P(J_{n+1} = i_{n+1} - i_n, I_{n+1} = 0 \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) \\
 &\quad + P(J_{n+1} = i_{n+1} - i_n + 1, I_{n+1} = -1 \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) \\
 &= P(J_{n+1} = i_{n+1} - i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n, I_{n+1} = 0) \\
 &\quad \cdot P(I_{n+1} = 0 \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) \\
 &\quad + P(J_{n+1} = i_{n+1} - i_n + 1 \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n, I_{n+1} = -1) \\
 &\quad \cdot P(I_{n+1} = -1 \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) \\
 &= P(J_{n+1} = i_{n+1} - i_n)(1 - \frac{i_n}{N}) + P(J_{n+1} = i_{n+1} - i_n + 1)\frac{i_n}{N} \\
 &= \begin{cases} p(1 - \frac{i_n}{N}), & i_{n+1} - i_n = 1, \\ q(1 - \frac{i_n}{N}) + p\frac{i_n}{N}, & i_{n+1} - i_n = 0, \\ q\frac{i_n}{N}, & i_{n+1} - i_n = -1, \\ 0, & \text{其他.} \end{cases} \tag{4}
 \end{aligned}$$

同理可得

$$P(X_{n+1} = i_{n+1} \mid X_n = i_n) = \begin{cases} p(1 - \frac{i_n}{N}), & i_{n+1} - i_n = 1, \\ q(1 - \frac{i_n}{N}) + p\frac{i_n}{N}, & i_{n+1} - i_n = 0, \\ q\frac{i_n}{N}, & i_{n+1} - i_n = -1, \\ 0, & \text{其他.} \end{cases} \tag{5}$$

由此及 (4) 可得

$$\begin{aligned}
 & P(X_{n+1} = i_{n+1} \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) \\
 &= P(X_{n+1} = i_{n+1} \mid X_n = i_n) \\
 &= \begin{cases} p(1 - \frac{i_n}{N}), & i_{n+1} - i_n = 1, \\ q(1 - \frac{i_n}{N}) + p\frac{i_n}{N}, & i_{n+1} - i_n = 0, \\ q\frac{i_n}{N}, & i_{n+1} - i_n = -1, \\ 0, & \text{其他.} \end{cases} \tag{6}
 \end{aligned}$$

这说明了, $\{X_n, n = 0, 1, 2, \dots, \}$ 为一 Markov 链, 且其转移概率矩阵为

$$P = \frac{1}{N} \begin{pmatrix} qN & pN & 0 & \cdots & 0 & 0 & 0 \\ q & q(N-1)+p & p(N-1) & \cdots & 0 & 0 & 0 \\ 0 & 2q & q(N-2)+2p & \cdots & 0 & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 2q+p(N-2) & 2p & 0 \\ 0 & 0 & 0 & \cdots & q(N-1) & q+p(N-1) & p \\ 0 & 0 & 0 & \cdots & 0 & qN & pN \end{pmatrix}.$$

■

5. 重复掷币一直到连续出现两次正面为止. 假定出现正面的概率是 p , 出现反面的概率是 q , $p+q=1$. 试引入以连续出现次数为状态空间的 Markov 链, 并求平均需要掷多少次实验才会停止.

解 以 I_n 表示第 n 次掷币的结果, 约定

$$I_n = \begin{cases} 1, & \text{第 } n \text{ 次掷币出现正面,} \\ 0, & \text{第 } n \text{ 次掷币出现反面,} \end{cases} \quad n = 1, 2, \dots, \quad (1)$$

则 $\{I_n, n = 1, 2, \dots\}$ 为一独立同分布随机变量序列, $P(I_1 = 1) = p, P(I_1 = 0) = q$. 令二维随机向量

$$X_n = (I_n, I_{n+1}), \quad n = 1, 2, \dots, \quad (2)$$

即 X_n 表示相邻第 n 和 $n+1$ 次掷币的结果, $n = 1, 2, \dots$. 往证 $\{X_n, n = 1, 2, \dots\}$ 是一状态空间为 $\mathcal{X} = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$ 的 Markov 链. 为此, 注意到对任意正整数 n 及任意状态 $i_k = (i_{k1}, i_{k2}) \in \mathcal{X}, k = 1, \dots, n$, 有

$$\begin{aligned} & P(X_k = (i_{k1}, i_{k2}), k = 1, \dots, n) \\ &= P(I_k = i_{k1}, k = 1, \dots, n, I_{n+1} = i_{n2}) \\ &= \prod_{k=1}^n P(I_k = i_{k1}) \cdot P(I_{n+1} = i_{n2}) \\ &= \prod_{k=1}^n (p^{i_{k1}} q^{1-i_{k1}}) \cdot p^{i_{n2}} q^{1-i_{n2}} \\ &= \begin{cases} p^{\sum_{k=1}^n i_{k1} + i_{n2}} q^{n+1 - (\sum_{k=1}^n i_{k1} + i_{n2})}, & i_{k2} = i_{k+1,1}, k = 1, \dots, n-1, \\ 0, & \text{其他,} \end{cases} \quad (3) \end{aligned}$$

因而, 对任意正整数 n 及任意状态 $i_k = (i_{k1}, i_{k2}) \in \mathcal{X}, k = 1, \dots, n+1$, 有

$$\begin{aligned} & P(X_{n+1} = (i_{n+1,1}, i_{n+1,2}) \mid X_k = (i_{k1}, i_{k2}), k = 1, \dots, n) \\ &= \frac{P(X_k = (i_{k1}, i_{k2}), k = 1, \dots, n+1)}{P(X_k = (i_{k1}, i_{k2}), k = 1, \dots, n)} \\ &= \begin{cases} p^{i_{n+1,2}} q^{1-i_{n+1,2}}, & i_{k2} = i_{k+1,1}, k = 1, \dots, n, \\ 0, & \text{其他.} \end{cases} \quad (4) \end{aligned}$$

同理可知, 对任意正整数 n 及任意状态 $i_k = (i_{k1}, i_{k2}) \in \mathcal{X}, k = n, n+1$, 有

$$\begin{aligned} & P(X_{n+1} = (i_{n+1,1}, i_{n+1,2}) \mid X_n = (i_{n1}, i_{n2})) \\ &= \begin{cases} p^{i_{n+1,2}} q^{1-i_{n+1,2}}, & i_{n2} = i_{n+1,1}, \\ 0, & \text{其他.} \end{cases} \end{aligned}$$

由此及 (4) 可知, 对任意正整数 n 及任意状态 $i_k = (i_{k1}, i_{k2}) \in \mathcal{X}, k = 1, \dots, n+1$, 有

$$\begin{aligned} & P(X_{n+1} = (i_{n+1,1}, i_{n+1,2}) \mid X_k = (i_{k1}, i_{k2}), k = 1, \dots, n) \\ &= P(X_{n+1} = (i_{n+1,1}, i_{n+1,2}) \mid X_n = (i_{n1}, i_{n2})) \\ &= \begin{cases} p^{i_{n+1,2}} q^{1-i_{n+1,2}}, & i_{k2} = i_{k+1,1}, k = 1, \dots, n, \\ 0, & \text{其他.} \end{cases} \end{aligned}$$

这说明了, $\{X_n, n = 1, 2, \dots\}$ 是一 markov 链, 且其转移概率矩阵为

$$P = \begin{pmatrix} q & p & 0 & 0 \\ 0 & 0 & q & p \\ q & p & 0 & 0 \\ 0 & 0 & q & p \end{pmatrix} \begin{matrix} (0,0) \\ (0,1) \\ (1,0) \\ (1,1) \end{matrix}. \quad (5)$$

而 $\{X_n, n = 1, 2, \dots\}$ 的初始分布为

$$P(X_1 = (0,0)) = q^2, \quad P(X_1 = (0,1)) = P(X_1 = (1,0)) = pq, \quad P(X_1 = (1,1)) = p^2. \quad (6)$$

以 T 表示在掷币过程中首次连续出现两次正面所需掷币的次数, 即

$$\{T = n\} = \{X_k \neq (1,1), k = 1, \dots, n-2, X_{n-1} = (1,1)\}, \quad n = 2, 3, \dots, \quad (7)$$

因而

$$P(T = 2) = P(X_1 = (1,1)) = p^2, \quad (8)$$

$$\begin{aligned} P(T = n) &= P(X_k \neq (1,1), k = 1, \dots, n-2, X_{n-1} = (1,1)) \\ &= \sum_{i_k \in \mathcal{X} - \{(1,1)\}} P(i_k = i_k, k = 1, \dots, n-2, X_n = (1,1)) \\ &= \sum_{i_k \in \mathcal{X} - \{(1,1)\}} P(X_1 = i_1) P_{i_1 i_2} \cdots P_{i_{n-3} i_{n-2}} P_{i_{n-2}, (1,1)}, n = 3, 4, \dots. \end{aligned} \quad (9)$$

(9) 说明了, 当 $n = 3, 4, \dots$ 时, $P(T = n)$ 是三维行向量

$$(q^2, pq, pq) \begin{pmatrix} q & p & 0 \\ 0 & 0 & q \\ q & p & 0 \end{pmatrix}^{n-3} \begin{pmatrix} q & p & 0 & 0 \\ 0 & 0 & q & p \\ q & p & 0 & 0 \end{pmatrix}$$

的最后一个元素. 而

$$(q^2, pq, pq) \begin{pmatrix} q & p & 0 & 0 \\ 0 & 0 & q & p \\ q & p & 0 & 0 \end{pmatrix} = (q^2, pq, pq^2, p^2q),$$

因而

$$P(T=3) = p^2q. \quad (10)$$

注意到矩阵

$$\begin{pmatrix} q & p & 0 \\ 0 & 0 & q \\ q & p & 0 \end{pmatrix}$$

的 Jordan 分解为

$$\begin{aligned} \begin{pmatrix} q & p & 0 \\ 0 & 0 & q \\ q & p & 0 \end{pmatrix} &= \frac{1}{pq(\lambda_2 - \lambda_3)} \begin{pmatrix} p & \lambda_2 & \lambda_3 \\ -q & q & q \\ 0 & \lambda_2 & \lambda_3 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \\ &\cdot \begin{pmatrix} q(\lambda_2 - \lambda_3) & 0 & q(\lambda_3 - \lambda_2) \\ -q\lambda_3 & -p\lambda_3 & q(p + \lambda_3) \\ q\lambda_2 & p\lambda_2 & -q(p + \lambda_2) \end{pmatrix}, \end{aligned} \quad (11)$$

其中 $\lambda_2 = \frac{q+\sqrt{q^2+4pq}}{2} \in (0, 1), \lambda_3 = \frac{q-\sqrt{q^2+4pq}}{2} \in (-1, 0)$, 因而有

$$\begin{aligned} \begin{pmatrix} q & p & 0 \\ 0 & 0 & q \\ q & p & 0 \end{pmatrix}^n &= \frac{1}{pq(\lambda_2 - \lambda_3)} \begin{pmatrix} p & \lambda_2 & \lambda_3 \\ -q & q & q \\ 0 & \lambda_2 & \lambda_3 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & \lambda_2^n & 0 \\ 0 & 0 & \lambda_3^n \end{pmatrix} \\ &\cdot \begin{pmatrix} q(\lambda_2 - \lambda_3) & 0 & q(\lambda_3 - \lambda_2) \\ -q\lambda_3 & -p\lambda_3 & q(p + \lambda_3) \\ q\lambda_2 & p\lambda_2 & -q(p + \lambda_2) \end{pmatrix} \\ &= \frac{1}{\lambda_2 - \lambda_3} \begin{pmatrix} q(\lambda_2^n - \lambda_3^n) & p(\lambda_2^n - \lambda_3^n) & pq(\lambda_2^{n-1} - \lambda_3^{n-1}) \\ q^2(\lambda_2^{n-1} - \lambda_3^{n-1}) & pq(\lambda_2^{n-1} - \lambda_3^{n-1}) & pq^2(\lambda_2^{n-2} - \lambda_3^{n-2}) \\ q(\lambda_2^n - \lambda_3^n) & p(\lambda_2^n - \lambda_3^n) & pq(\lambda_2^{n-1} - \lambda_3^{n-1}) \end{pmatrix}, \quad n = 1, 2, \dots \end{aligned} \quad (12)$$

故

$$\begin{aligned} (q^2, pq, pq) \begin{pmatrix} q & p & 0 \\ 0 & 0 & q \\ q & p & 0 \end{pmatrix}^n \begin{pmatrix} q & p & 0 & 0 \\ 0 & 0 & q & p \\ q & p & 0 & 0 \end{pmatrix} \\ = \frac{q}{\lambda_2 - \lambda_3} (q^2[pq(1+p)(\lambda_2^{n-2} - \lambda_3^{n-2}) + (1+pq)(\lambda_2^{n-1} - \lambda_3^{n-1})]), \end{aligned}$$

$$\begin{aligned}
& pq[pq(1+p)(\lambda_2^{n-2} - \lambda_3^{n-2}) + (1+pq)(\lambda_2^{n-1} - \lambda_3^{n-1})], \\
& pq[\lambda_2^n - \lambda_3^n + pq(\lambda_2^{n-1} - \lambda_3^{n-1})], \\
& p^2[\lambda_2^n - \lambda_3^n + pq(\lambda_2^{n-1} - \lambda_3^{n-1})], \quad n = 1, 2, \dots
\end{aligned}$$

由上所述, 有

$$P(T = n) = \frac{p^2 q}{\lambda_2 - \lambda_3} [\lambda_2^{n-3} - \lambda_3^{n-3} + pq(\lambda_2^{n-4} - \lambda_3^{n-4})], \quad n = 4, 5, \dots \quad (13)$$

由 (8), (10) 和 (13) 可知, 为了在掷币过程中连续出现两次正面, 所需掷币的平均次数为

$$\begin{aligned}
E(T) &= \sum_{n=2}^{\infty} nP(T = n) \\
&= 2p^2 + 3p^2 q + \frac{p^2 q}{\lambda_2 - \lambda_3} \sum_{n=4}^{\infty} n[\lambda_2^{n-3} - \lambda_3^{n-3} + pq(\lambda_2^{n-4} - \lambda_3^{n-4})] \\
&= \frac{1+p}{p^2},
\end{aligned}$$

在上面的求和中, 用到了以下等式

$$\sum_{n=m}^{\infty} nx^{n-1} = \frac{x^{m-1}[m - (m-1)x]}{(1-x)^2}, \quad |x| < 1, m = 1, 2, \dots$$

■

6. 迷宫问题. 将小鼠放入迷宫中作动物的学习实验, 如图 3.3 所示. 在迷宫的第 7 号小格内放有美味食物而第 8 号小格内则是电击捕鼠装置. 假定当家鼠位于某格时有 k 个出口可以离去, 则它总是随机地选择一个, 概率为 $\frac{1}{k}$, 并假定每一次家鼠只能跑到相邻的小格去. 令 X_n 为家鼠在时刻 n 时所在小格的号数, $n = 0, 1, 2, \dots$. 试写出过程 $\{X_n, n = 0, 1, 2, \dots\}$ 的转移概率矩阵, 并求出家鼠在遭到电击前能找到食物的概率.

解 由题设可知, $\{X_n, n = 0, 1, 2, \dots\}$ 是一 Markov 链, 其转移概率矩阵为

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & 0 & 0 & \frac{1}{3} & 0 \\ \frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 0 & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \end{matrix} \quad (1)$$

以 $f_{i7}^{(n)}$ 表示家鼠从 $X_0 = i$ 出发经 n 步未受电击首次找到食物的概率, 即

$$f_{i7}^{(n)} = P(X_n = 7, X_k \neq 7, 8, k = 1, \dots, n-1 \mid X_0 = i), \quad n = 1, 2, \dots, \quad (2)$$

则

$$f_{i7}^{(1)} = P_{i7}, \quad i = 0, 1, \dots, 8,$$

即

$$\begin{cases} f_{07}^{(1)} = f_{27}^{(1)} = f_{37}^{(1)} = f_{57}^{(1)} = f_{67}^{(1)} = f_{77}^{(1)} = f_{87}^{(1)} = 0 \\ f_{17}^{(1)} = f_{47}^{(1)} = \frac{1}{3}. \end{cases} \quad (3)$$

而对 $n = 2, 3, \dots$, $f_{i7}^{(n)}$ 是

$$P_{[7,8]} P_{(7,8)[7,8]}^{n-2} P_{(7,8)} \quad (4)$$

的状态 i 所在行, 状态 7 所在列交叉位置上的元素, $i = 0, 1, \dots, 8$, 其中 $P_{(7,8)}$ 表示从 P 中删除状态 7, 8 所在行得到的 7×9 矩阵, $P_{[7,8]}$ 表示从 P 中删除状态 7, 8 所在列得到的 9×7 矩阵, $P_{(7,8),[7,8]}$ 表示从 P 中删除状态 7, 8 所在行和列得到的 7×7 矩阵.

注意到

$$P_{[7,8]} P_{(7,8)} = \begin{pmatrix} \frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & 0 & 0 & \frac{1}{6} & \frac{1}{6} \\ 0 & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{12} & \frac{1}{12} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{12} & \frac{1}{12} & 0 & 0 & 0 \\ \frac{1}{6} & 0 & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & \frac{1}{12} & \frac{1}{12} & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{12} & \frac{1}{12} & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{3} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & 0 & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{6} & \frac{1}{3} & 0 \\ \frac{1}{6} & 0 & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{6} & 0 & \frac{1}{3} \end{pmatrix},$$

因此由上所述得

$$\begin{cases} f_{07}^{(2)} = f_{37}^{(2)} = f_{67}^{(2)} = \frac{1}{6}, \\ f_{17}^{(2)} = f_{27}^{(2)} = f_{47}^{(2)} = f_{57}^{(2)} = f_{87}^{(2)} = 0, \\ f_{77}^{(2)} = \frac{1}{3}. \end{cases} \quad (5)$$

进而, 注意到 $P_{(7,8),[7,8]}$ 的 Jordan 分解为

$$P_{(7,8),[7,8]} = T J T^{-1}, \quad (6)$$

其中

$$T = \begin{pmatrix} 1 & 0 & 0 & \lambda_1 & \lambda_2 & -\lambda_3 & \lambda_4 \\ 0 & 1 & 0 & 1 & 1 & -1 & 1 \\ 0 & -1 & 0 & 1 & 1 & -1 & 1 \\ -1 & 0 & 0 & \lambda_1 & \lambda_2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & -1 \\ 0 & 0 & -1 & 1 & 1 & 1 & -1 \\ 1 & 0 & 0 & \lambda_1 & \lambda_2 & \lambda_3 & -\lambda_4 \end{pmatrix}, \quad (7)$$

$$J = \text{diag}(0, 0, 0, \lambda_1, \lambda_2, \lambda_3, \lambda_4), \quad (8)$$

$$T^{-1} = \begin{pmatrix} \frac{1}{4} & 0 & 0 & -\frac{1}{2} & 0 & 0 & \frac{1}{4} \\ 0 & \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ \frac{\lambda_1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{\lambda_1}{4} & \frac{1}{8} & \frac{1}{8} & \frac{\lambda_1}{8} \\ \frac{\lambda_2}{8} & \frac{1}{8} & \frac{1}{8} & \frac{\lambda_2}{4} & \frac{1}{8} & \frac{1}{8} & \frac{\lambda_2}{8} \\ -\frac{\lambda_3}{4} & -\frac{1}{8} & -\frac{1}{8} & 0 & \frac{1}{8} & \frac{1}{8} & \frac{\lambda_3}{4} \\ \frac{\lambda_4}{4} & \frac{1}{8} & \frac{1}{8} & 0 & -\frac{1}{8} & -\frac{1}{8} & -\frac{\lambda_4}{4} \end{pmatrix}, \quad (9)$$

其中 $\lambda_{1,2} = \pm\sqrt{\frac{2}{3}}, \lambda_{3,4} = \pm\frac{1}{\sqrt{3}}$, 因此

$$\begin{aligned} P_{[7,8]} P_{(7,8),[7,8]}^n P_{(7,8)} &= P_{[7,8]} T J^n T^{-1} P_{(7,8)} \\ &= \begin{pmatrix} \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_3^n + \lambda_4^n) & \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1} + \lambda_1 \lambda_3^n + \lambda_2 \lambda_4^n) \\ \frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1} + \lambda_3^{n-1} + \lambda_4^{n-1}) & \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_1 \lambda_3^{n-1} + \lambda_2 \lambda_4^{n-1}) \\ \frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1} + \lambda_3^{n-1} + \lambda_4^{n-1}) & \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_1 \lambda_3^{n-1} + \lambda_2 \lambda_4^{n-1}) \\ \frac{1}{12}(\lambda_1^n + \lambda_2^n) & \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1}) \\ \frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1} - \lambda_3^{n-1} - \lambda_4^{n-1}) & \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_2 \lambda_3^{n-1} + \lambda_1 \lambda_4^{n-1}) \\ \frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1} - \lambda_3^{n-1} - \lambda_4^{n-1}) & \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_2 \lambda_3^{n-1} + \lambda_1 \lambda_4^{n-1}) \\ \frac{1}{12}(\lambda_1^n + \lambda_2^n - \lambda_3^n - \lambda_4^n) & \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1} + \lambda_2 \lambda_3^n + \lambda_1 \lambda_4^n) \\ \frac{1}{12}(\lambda_1^n + \lambda_2^n) & \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1}) \\ \frac{1}{12}(\lambda_1^n + \lambda_2^n) & \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1}) \end{pmatrix} \\ &\quad \begin{pmatrix} \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1} + \lambda_1 \lambda_3^n + \lambda_2 \lambda_4^n) & \frac{1}{6}(\lambda_1^n + \lambda_2^n) \\ \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_1 \lambda_3^{n-1} + \lambda_2 \lambda_4^{n-1}) & \frac{1}{9}(\lambda_1^{n-1} + \lambda_2^{n-1}) \\ \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_1 \lambda_3^{n-1} + \lambda_2 \lambda_4^{n-1}) & \frac{1}{9}(\lambda_1^{n-1} + \lambda_2^{n-1}) \\ \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1}) & \frac{1}{6}(\lambda_1^n + \lambda_2^n) \\ \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_2 \lambda_3^{n-1} + \lambda_1 \lambda_4^{n-1}) & \frac{1}{9}(\lambda_1^{n-1} + \lambda_2^{n-1}) \\ \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_2 \lambda_3^{n-1} + \lambda_1 \lambda_4^{n-1}) & \frac{1}{9}(\lambda_1^{n-1} + \lambda_2^{n-1}) \\ \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1} + \lambda_2 \lambda_3^n + \lambda_1 \lambda_4^n) & \frac{1}{6}(\lambda_1^n + \lambda_2^n) \\ \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1}) & \frac{1}{6}(\lambda_1^n + \lambda_2^n) \\ \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1}) & \frac{1}{6}(\lambda_1^n + \lambda_2^n) \end{pmatrix} \end{aligned}$$

$$\begin{pmatrix}
\frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1} + \lambda_2\lambda_3^n + \lambda_1\lambda_4^n) & \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1} + \lambda_2\lambda_3^n + \lambda_1\lambda_4^n) \\
\frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_2\lambda_3^{n-1} + \lambda_1\lambda_4^{n-1}) & \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_2\lambda_3^{n-1} + \lambda_1\lambda_4^{n-1}) \\
\frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_2\lambda_3^{n-1} + \lambda_1\lambda_4^{n-1}) & \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_2\lambda_3^{n-1} + \lambda_1\lambda_4^{n-1}) \\
\frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1}) & \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1}) \\
\frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_1\lambda_3^{n-1} + \lambda_2\lambda_4^{n-1}) & \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_1\lambda_3^{n-1} + \lambda_2\lambda_4^{n-1}) \\
\frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_1\lambda_3^{n-1} + \lambda_2\lambda_4^{n-1}) & \frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_1\lambda_3^{n-1} + \lambda_2\lambda_4^{n-1}) \\
\frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1} + \lambda_1\lambda_3^n + \lambda_2\lambda_4^n) & \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1} + \lambda_1\lambda_3^n + \lambda_2\lambda_4^n) \\
\frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1}) & \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1}) \\
\frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1}) & \frac{1}{8}(\lambda_1^{n+1} + \lambda_2^{n+1}) \\
\frac{1}{12}(\lambda_1^n + \lambda_2^n - \lambda_3^n - \lambda_4^n) & \frac{1}{12}(\lambda_1^n + \lambda_2^n) & \frac{1}{12}(\lambda_1^n + \lambda_2^n) \\
\frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1} - \lambda_3^{n-1} - \lambda_4^{n-1}) & \frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1}) & \frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1}) \\
\frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1} - \lambda_3^{n-1} - \lambda_4^{n-1}) & \frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1}) & \frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1}) \\
\frac{1}{12}(\lambda_1^n + \lambda_2^n) & \frac{1}{12}(\lambda_1^n + \lambda_2^n) & \frac{1}{12}(\lambda_1^n + \lambda_2^n) \\
\frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1} + \lambda_3^{n-1} + \lambda_4^{n-1}) & \frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1}) & \frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1}) \\
\frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1} + \lambda_3^{n-1} + \lambda_4^{n-1}) & \frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1}) & \frac{1}{18}(\lambda_1^{n-1} + \lambda_2^{n-1}) \\
\frac{1}{12}(\lambda_1^n + \lambda_2^n + \lambda_3^n + \lambda_4^n) & \frac{1}{12}(\lambda_1^n + \lambda_2^n) & \frac{1}{12}(\lambda_1^n + \lambda_2^n) \\
\frac{1}{12}(\lambda_1^n + \lambda_2^n) & \frac{1}{12}(\lambda_1^n + \lambda_2^n) & \frac{1}{12}(\lambda_1^n + \lambda_2^n) \\
\frac{1}{12}(\lambda_1^n + \lambda_2^n) & \frac{1}{12}(\lambda_1^n + \lambda_2^n) & \frac{1}{12}(\lambda_1^n + \lambda_2^n)
\end{pmatrix}, \quad (10)$$

因而由上所述得

$$\begin{cases} f_{07}^{(n)} = f_{37}^{(n)} = f_{67}^{(n)} = f_{77}^{(n)} = f_{87}^{(n)} = \frac{1}{12}(\lambda_1^{n-2} + \lambda_2^{n-2}), \\ f_{17}^{(n)} = f_{27}^{(n)} = f_{47}^{(n)} = f_{57}^{(n)} = \frac{1}{18}(\lambda_1^{n-3} + \lambda_2^{n-3}), \end{cases} \quad n = 3, 4, \dots \quad (11)$$

若以 f_{i7} 表示家鼠从 $X_0 = i$ 出发未受电击找到食物的概率, 则

$$f_{i7} = \sum_{n=1}^{\infty} f_{i7}^{(n)}, \quad i = 0, 1, \dots, 8. \quad (12)$$

由 (3), (5) 和 (11) 易求得

$$\begin{cases} f_{07} = f_{37} = f_{67} = \frac{1}{2}, \\ f_{17} = f_{47} = f_{77} = \frac{2}{3}, \\ f_{27} = f_{57} = f_{87} = \frac{1}{3}. \end{cases} \quad (13)$$

■

7. 设 $Z_i, i = 1, 2, \dots$ 是一串独立同分布的离散随机变量, 分布为 $P(Z_n = k) = p_k, k = 0, 1, 2, \dots, \sum_{k=0}^{\infty} p_k = 1$. 试证 $\{Z_n, n = 1, 2, \dots\}$ 是一 Markov 链, 并求其转移概率矩阵.

解证 由题设知, $\{Z_n, n = 1, 2, \dots\}$ 的状态空间为 $\mathcal{X} = \{0, 1, 2, \dots\}$, 对任意正整数 n 及任意 $i_1, \dots, i_{n+1} \in \mathcal{X}$, 有

$$P(Z_{n+1} = i_{n+1} \mid Z_1 = i_1, \dots, Z_n = i_n) = P(Z_{n+1} = i_{n+1}) = p_{i_{n+1}}$$

和

$$P(Z_{n+1} = i_{n+1} \mid Z_n = i_n) = P(Z_{n+1} = i_{n+1}) = p_{i_{n+1}}.$$

因而

$$P(Z_{n+1} = i_{n+1} \mid Z_1 = i_1, \dots, Z_n = i_n) = P(Z_{n+1} = i_{n+1} \mid Z_n = i_n) = p_{i_{n+1}}.$$

这说明了 $\{Z_n, n = 1, 2, \dots\}$ 是一 Markov 链, 且其转移概率矩阵为

$$P = \begin{pmatrix} p_0 & p_1 & p_2 & \cdots \\ p_0 & p_1 & p_2 & \cdots \\ p_0 & p_1 & p_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

■

8. 对第 7 题中的 $\{Z_i, i = 1, 2, \dots\}$, 令 $X_n = \max\{Z_1, \dots, Z_n\}, n = 1, 2, \dots$, 并约定 $X_0 \equiv 0$. 试问 $\{X_n, n = 0, 1, 2, \dots\}$ 是否为 Markov 链? 若是, 则求其转移概率矩阵.

解 由题设知, $\{X_n, n = 0, 1, 2, \dots\}$ 的状态空间为 $\mathcal{X} = \{0, 1, 2, \dots\}$. 由于 $P(X_0 = 0) = 1$, 因此, $X_0, Z_1, Z_2, \dots$ 独立. 对任意 $i_1 \in \mathcal{X}$, 有

$$P(X_1 = i_1 \mid X_0 = 0) = P(Z_1 = i_1 \mid X_0 = 0) = P(Z_1 = i_1) = p_{i_1}. \quad (1)$$

进而, 对任意正整数 n 及任意 $i_1, \dots, i_{n+1} \in \mathcal{X}$, 有

$$\begin{aligned} & P(X_{n+1} = i_{n+1} \mid X_0 = 0, X_1 = i_1, \dots, X_n = i_n) \\ &= \frac{P(X_0 = 0, X_1 = i_1, \dots, X_n = i_n, X_{n+1} = i_{n+1})}{P(X_0 = 0, X_1 = i_1, \dots, X_n = i_n)} \\ &= \frac{P(X_1 = i_1, \dots, X_n = i_n, X_{n+1} = i_{n+1})}{P(X_1 = i_1, \dots, X_n = i_n)} \\ &= \frac{P(Z_1 = i_1, \max\{i_1, Z_2\} = i_2, \dots, \max\{i_{n-1}, Z_n\} = i_n, \max\{i_n, Z_{n+1}\} = i_{n+1})}{P(Z_1 = i_1, \max\{i_1, Z_2\} = i_2, \dots, \max\{i_{n-1}, Z_n\} = i_n)} \\ &= P(\max\{i_n, Z_{n+1}\} = i_{n+1}) \\ &= \begin{cases} 0, & i_n > i_{n+1}, \\ P(Z_{n+1} \leq i_{n+1}), & i_n = i_{n+1}, \\ P(\max\{i_n, Z_{n+1}\} = i_{n+1}), & i_n < i_{n+1} \end{cases} \\ &= \begin{cases} 0, & i_n > i_{n+1}, \\ P(Z_{n+1} \leq i_{n+1}), & i_n = i_{n+1}, \\ P(\max\{i_n, Z_{n+1}\} = i_{n+1}, Z_{n+1} < i_n) \\ \quad + P(\max\{i_n, Z_{n+1}\} = i_{n+1}, Z_{n+1} \geq i_n), & i_n < i_{n+1} \end{cases} \\ &= \begin{cases} 0, & i_n > i_{n+1}, \\ P(Z_{n+1} \leq i_{n+1}), & i_n = i_{n+1}, \\ P(Z_{n+1} = i_{n+1}), & i_n < i_{n+1} \end{cases} \end{aligned}$$

$$= \begin{cases} 0, & i_n > i_{n+1}, \\ \sum_{k=0}^{i_{n+1}} p_k, & i_n = i_{n+1}, \\ p_{i_{n+1}}, & i_n < i_{n+1}. \end{cases} \quad (2)$$

同理可知, 对任意正整数 n 及任意 $i_n, i_{n+1} \in \mathcal{X}$, 有

$$P(X_{n+1} = i_{n+1} \mid X_n = i_n) = \begin{cases} 0, & i_n > i_{n+1}, \\ \sum_{k=0}^{i_{n+1}} p_k, & i_n = i_{n+1}, \\ p_{i_{n+1}}, & i_n < i_{n+1}. \end{cases} \quad (3)$$

由此及 (2) 可得

$$\begin{aligned} & P(X_{n+1} = i_{n+1} \mid X_0 = 0, X_1 = i_1, \dots, X_n = i_n) \\ &= P(X_{n+1} = i_{n+1} \mid X_n = i_n). \end{aligned} \quad (4)$$

由此及 (1) 可知, $\{X_n, n = 0, 1, 2, \dots\}$ 为一 Markov 链, 且其转移概率矩阵为

$$P = \begin{pmatrix} p_0 & p_1 & p_2 & p_3 & \cdots \\ 0 & p_0 + p_1 & p_2 & p_3 & \cdots \\ 0 & 0 & p_0 + p_1 + p_2 & p_3 & \cdots \\ 0 & 0 & 0 & p_0 + p_1 + p_2 + p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

■

10. 对第 7 题中的 $Z_i, i = 1, 2, \dots$, 若定义 $X_n = \sum_{i=1}^n Z_i, n = 1, 2, \dots, X_0 \equiv 0$, 试证 $\{X_n, n = 0, 1, 2, \dots\}$ 为一 Markov 链, 并求其转移概率矩阵.

解 由题设可知, 对任意状态 $i_1 \in \mathcal{X}$, 有

$$P(X_1 = i_1 \mid X_0 = 0) = P(X_1 = i_1) = p_{i_1}. \quad (1)$$

进而, 对任意正整数 n 及任意状态 $i_k, k = 1, \dots, n+1$, 有

$$\begin{aligned} & P(X_{n+1} = i_{n+1} \mid X_0 = 0, X_k = i_k, k = 1, \dots, n) \\ &= P(Z_{n+1} = i_{n+1} - i_n \mid X_k = i_k, k = 1, \dots, n) \\ &= P(Z_{n+1} = i_{n+1} - i_n) \\ &= \begin{cases} p_{i_{n+1}-i_n}, & i_{n+1} - i_n = 0, 1, 2, \dots, \\ 0, & \text{其他.} \end{cases} \end{aligned} \quad (2)$$

同理可知, 对任意正整数 n 及任意状态 i_n, i_{n+1} , 有

$$\begin{aligned} & P(X_{n+1} = i_{n+1} \mid X_n = i_n) = P(Z_{n+1} = i_{n+1} - i_n) \\ &= \begin{cases} p_{i_{n+1}-i_n}, & i_{n+1} - i_n = 0, 1, 2, \dots, \\ 0, & \text{其他.} \end{cases} \end{aligned} \quad (3)$$

由 (1),(2) 和 (3) 即知, $\{X_n, n = 0, 1, 2, \dots\}$ 为一 Markov 链, 且其转移概率矩阵为

$$P = \begin{pmatrix} p_0 & p_1 & p_2 & p_3 & \cdots \\ 0 & p_0 & p_1 & p_2 & \cdots \\ 0 & 0 & p_0 & p_1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

■

11. 一 Markov 链有状态 0,1,2,3 和转移概率矩阵

$$P = \begin{pmatrix} 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix},$$

试求 $f_{00}^{(n)}, n = 1, 2, \dots$ 及 f_{00} .

解 首先, $f_{00}^{(1)} = P_{00} = 0$. 其次, 对 $n \geq 2$, $f_{00}^{(n)}$ 是矩阵

$$\begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}^{n-2} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix}$$

的 (1,1)-元, $n = 2, 3, \dots$. 而

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}^n = \begin{pmatrix} 0 & 0 & \frac{1}{2^{n-2}} \\ 0 & 0 & \frac{1}{2^{n-1}} \\ 0 & 0 & \frac{1}{2^n} \end{pmatrix}, \quad n = 2, 3, \dots,$$

因而

$$\begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & 0 & \frac{1}{2} & \frac{1}{4} \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{4} & 0 & 0 & \frac{1}{4} \end{pmatrix},$$

$$\begin{aligned} & \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{8} & 0 & 0 & \frac{5}{8} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{4} & 0 & 0 & \frac{1}{4} \\ \frac{1}{8} & 0 & 0 & \frac{1}{8} \end{pmatrix}, \end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}^n \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix} \\
&= \begin{pmatrix} \frac{5}{2^{n+2}} & 0 & 0 & \frac{5}{2^{n+2}} \\ \frac{1}{2^n} & 0 & 0 & \frac{1}{2^n} \\ \frac{1}{2^{n+1}} & 0 & 0 & \frac{1}{2^{n+1}} \\ \frac{1}{2^{n+2}} & 0 & 0 & \frac{1}{2^{n+2}} \end{pmatrix}, \quad n = 2, 3, \dots
\end{aligned}$$

故

$$f_{00}^{(2)} = \frac{1}{4}, \quad f_{00}^{(3)} = \frac{1}{8}, \quad f_{00}^{(n)} = \frac{5}{2^n}, \quad n = 4, 5, \dots$$

从而

$$f_{00} = \sum_{n=1}^{\infty} f_{00}^{(n)} = \frac{1}{4} + \frac{1}{8} + \sum_{n=4}^{\infty} \frac{5}{2^n} = 1.$$

■

12. 在成败型的重复试验中, 每次试验结果为成功 (S) 或失败 (F). 同一结果相继出现称为一个游程 (Run), 比如结果 FSSFFFSF 中共有两个成功游程和三个失败游程. 设成功概率为 p , 失败概率为 $q = 1 - p$. 记 X_n 为 n 次试验后成功游程的长度 (若 n 次试验失败则 $X_n = 0$), $n = 1, 2, \dots$. 试证 $\{X_n, n = 1, 2, \dots\}$ 为一 Markov 链, 并确定其转移概率阵. 记 T 为返回状态 0 的时间, 试求 T 的分布及均值, 并由此对这一 Markov 链的状态进行分类.

解证 若以 I_n 表示第 n 次试验的结果, 约定

$$I_n = \begin{cases} 1, & \text{第 } n \text{ 次试验成功,} \\ 0, & \text{第 } n \text{ 次试验失败,} \end{cases} \quad n = 1, 2, \dots, \quad (1)$$

则 $\{I_n, n = 1, 2, \dots\}$ 为一独立同分布随机变量序列, $P(I_n = 1) = p$, $P(I_n = 0) = q$, $n = 1, 2, \dots$, 且

$$X_n = \sum_{k=1}^n I_k, \quad n = 1, 2, \dots. \quad (2)$$

往证 $\{X_n, n = 1, 2, \dots\}$ 是状态空间为 $\mathcal{X} = \{0, 1, 2, \dots\}$ 的 Markov 链. 其实, 对任意正整数 n 及任意状态 $i_1, \dots, i_{n+1} \in \mathcal{X}$, 有

$$\begin{aligned}
& P(X_{n+1} = i_{n+1} \mid X_1 = i_1, \dots, X_n = i_n) \\
&= P(I_{n+1} = i_{n+1} - i_n \mid X_1 = i_1, \dots, X_n = i_n) \\
&= P(I_{n+1} = i_{n+1} - i_n) \\
&= \begin{cases} p^{i_{n+1}-i_n} q^{1-(i_{n+1}-i_n)}, & i_{n+1} - i_n = 0, 1, \\ 0, & \text{其他,} \end{cases} \quad i_k = 0, 1, \dots, k, k = 1, \dots, n+1.
\end{aligned}$$

这说明了 $P(X_{n+1} = i_{n+1} | X_1 = i_1, \dots, X_n = i_n)$ 只与 i_n 和 i_{n+1} 有关, 因而 $\{X_n, n = 1, 2, \dots\}$ 为 Markov 链, 且其转移概率阵为

$$P = \begin{pmatrix} q & p & 0 & 0 & 0 & \cdots \\ 0 & q & p & 0 & 0 & \cdots \\ 0 & 0 & q & p & 0 & \cdots \\ 0 & 0 & 0 & q & p & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \end{matrix} \quad (3)$$

往求 $f_{00}^{(n)}, n = 1, 2, \dots$. 首先, 有

$$f_{00}^{(1)} = P_{00} = q. \quad (4)$$

其次, $f_{00}^{(n)}$ 是

$$\begin{aligned} P_{[0]} P_{(0)[0]}^{n-2} P_{(0)} &= \begin{pmatrix} pe_1 \\ P \end{pmatrix} P^{n-2}(0, P) \\ &= \begin{pmatrix} 0 & pe_1 P^{n-1} \\ 0 & P^n \end{pmatrix} \end{aligned}$$

的状态 0 所在行, 状态 0 所在列交叉位置上的元素 $0, n = 2, 3, \dots$, 其中 $e_1 = (1, 0, 0, \dots)$, 即

$$f_{00}^{(n)} = 0, \quad n = 2, 3, \dots \quad (5)$$

由 (4) 和 (5) 可得返回状态 0 的时间 T 的分布为

$$\begin{cases} P(T = 1 | X_m = 0) = f_{00}^{(1)} = q, \\ P(T = n | X_m = 0) = f_{00}^{(n)} = 0, \quad n = 2, 3, \dots, \quad m = 1, 2, \dots \\ P(T = +\infty | X_m = 0) = p, \end{cases} \quad (6)$$

这说明了状态 0 是瞬过的, 且 $E(T | X_m = 0) = +\infty$. ■

16. 考虑一生长与灾害模型. 这类 Markov 链 $\{X_n, n = 0, 1, 2, \dots\}$ 有状态 $0, 1, 2, \dots$. 当过程处于状态 i 时既可能以概率 p_i 转移到状态 $i+1$ (生长), 也可能以概率 $q_i = 1 - p_i$ 落回状态 0 (灾害), $i = 1, 2, \dots$. 而从状态 0 又必然“无中生有”, 即 $P_{01} = 1$.

(a) 试证所有状态为常返的条件是 $\lim_{n \rightarrow \infty} (p_1 p_2 \cdots p_n) = 0$.

(b) 若此链是常返的, 试求其为零常返的条件.

解证 (a) 注意到 $\{X_n, n = 0, 1, 2, \dots\}$ 的转移概率阵为

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & \cdots \\ q_1 & 0 & p_1 & 0 & 0 & \cdots \\ q_2 & 0 & 0 & p_2 & 0 & \cdots \\ q_3 & 0 & 0 & 0 & p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \end{matrix} \quad (1)$$

由此可知, 在状态空间 $\mathcal{X} = \{0, 1, 2, \dots\}$ 中的任意两个状态 i 和 $j (> i)$ 依以下方式互达:

$$i \xrightarrow{p_i} i+1 \xrightarrow{p_{i+1}} \dots \xrightarrow{p_{j-1}} j \xrightarrow{q_j} 0 \xrightarrow{1} 1 \xrightarrow{p_1} \dots \xrightarrow{p_{i-1}} i. \quad (2)$$

因而, 为了证明所有状态均常返 $\Leftrightarrow \lim_{n \rightarrow \infty} (p_1 p_2 \cdots p_n) = 0$, 只需证明状态 0 常返 $\Leftrightarrow \lim_{n \rightarrow \infty} (p_1 p_2 \cdots p_n) = 0$. 为此往求 $f_{00}^{(n)}$, $n = 1, 2, \dots$. 首先, 有

$$f_{00}^{(1)} = P_{00} = 0. \quad (3)$$

其次, $f_{00}^{(2)}$ 是

$$\begin{aligned} P_{[0]} P_{(0)} &= \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots \\ 0 & p_1 & 0 & 0 & \cdots \\ 0 & 0 & p_2 & 0 & \cdots \\ 0 & 0 & 0 & p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} q_1 & 0 & p_1 & 0 & 0 & \cdots \\ q_2 & 0 & 0 & p_2 & 0 & \cdots \\ q_3 & 0 & 0 & 0 & p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \\ &= \begin{pmatrix} q_1 & 0 & p_1 & 0 & 0 & \cdots \\ p_1 q_2 & 0 & 0 & p_1 p_2 & 0 & \cdots \\ p_2 q_3 & 0 & 0 & 0 & p_2 p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \end{aligned}$$

的状态 0 所在行, 状态 0 所在列交叉位置上的元素, 即

$$f_{00}^{(2)} = q_1. \quad (4)$$

最后, 对 $n = 3, 4, \dots$, $f_{00}^{(n)}$ 是

$$\begin{aligned} &P_{[0]} P_{(0)[0]}^{n-2} P_{(0)} \\ &= \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots \\ 0 & p_1 & 0 & 0 & \cdots \\ 0 & 0 & p_2 & 0 & \cdots \\ 0 & 0 & 0 & p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} 0 & p_1 & 0 & 0 & \cdots \\ 0 & 0 & p_2 & 0 & \cdots \\ 0 & 0 & 0 & p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}^{n-2} \begin{pmatrix} q_1 & 0 & p_1 & 0 & 0 & \cdots \\ q_2 & 0 & 0 & p_2 & 0 & \cdots \\ q_3 & 0 & 0 & 0 & p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \end{aligned}$$

的状态 0 所在行, 状态 0 所在列交叉位置上的元素, $n = 3, 4, \dots$. 而

$$\begin{aligned} P_{(0)[0]}^n &= \begin{pmatrix} 0 & p_1 & 0 & 0 & \cdots \\ 0 & 0 & p_2 & 0 & \cdots \\ 0 & 0 & 0 & p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}^n \\ &= \begin{pmatrix} \underbrace{0 \cdots 0}_{n \uparrow} & p_1 \cdots p_n & 0 & 0 & 0 & \cdots \\ \underbrace{0 \cdots 0}_{n \uparrow} & 0 & p_2 \cdots p_{n+1} & 0 & 0 & \cdots \\ \underbrace{0 \cdots 0}_{n \uparrow} & 0 & 0 & p_3 \cdots p_{n+2} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad n = 1, 2, \dots, \end{aligned}$$

因此

$$\begin{aligned}
 & P_{[0]} P_{(0)[0]}^n P_{(0)} \\
 &= \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots \\ 0 & p_1 & 0 & 0 & \cdots \\ 0 & 0 & p_2 & 0 & \cdots \\ 0 & 0 & 0 & p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} \underbrace{0 \cdots 0}_{n \uparrow} & p_1 \cdots p_n & 0 & 0 & 0 & \cdots \\ \underbrace{0 \cdots 0}_{n \uparrow} & 0 & p_2 \cdots p_{n+1} & 0 & 0 & \cdots \\ \underbrace{0 \cdots 0}_{n \uparrow} & 0 & 0 & p_3 \cdots p_{n+2} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \\
 &\quad \cdot \begin{pmatrix} q_1 & 0 & p_1 & 0 & 0 & \cdots \\ q_2 & 0 & 0 & p_2 & 0 & \cdots \\ q_3 & 0 & 0 & 0 & p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \\
 &= \begin{pmatrix} \underbrace{0 \cdots 0}_{n \uparrow} & p_1 \cdots p_n & 0 & 0 & 0 & \cdots \\ \underbrace{0 \cdots 0}_{n \uparrow} & 0 & p_1 \cdots p_{n+1} & 0 & 0 & \cdots \\ \underbrace{0 \cdots 0}_{n \uparrow} & 0 & 0 & p_2 \cdots p_{n+2} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} q_1 & 0 & p_1 & 0 & 0 & \cdots \\ q_2 & 0 & 0 & p_2 & 0 & \cdots \\ q_3 & 0 & 0 & 0 & p_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \\
 &= \begin{pmatrix} p_1 \cdots p_n q_{n+1} & \underbrace{0 \cdots 0}_{n+1 \uparrow} & p_1 \cdots p_{n+1} & 0 & 0 & 0 & \cdots \\ p_1 \cdots p_{n+1} q_{n+2} & \underbrace{0 \cdots 0}_{n+1 \uparrow} & 0 & p_1 \cdots p_{n+2} & 0 & 0 & \cdots \\ p_2 \cdots p_{n+2} q_{n+3} & \underbrace{0 \cdots 0}_{n+1 \uparrow} & 0 & 0 & p_2 \cdots p_{n+3} & 0 & \cdots \\ p_3 \cdots p_{n+3} q_{n+4} & \underbrace{0 \cdots 0}_{n+1 \uparrow} & 0 & 0 & 0 & p_3 \cdots p_{n+4} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \\
 &\quad n = 1, 2, \cdots,
 \end{aligned}$$

故由上所述得

$$f_{00}^{(n)} = p_1 \cdots p_{n-2} q_{n-1}, \quad n = 3, 4, \cdots. \quad (5)$$

由 (3), (4) 和 (5) 可得

$$\begin{aligned}
 f_{00} &= \sum_{n=1}^{\infty} f_{00}^{(n)} = q_1 + \sum_{n=3}^{+\infty} (p_1 \cdots p_{n-2} q_{n-1}) \\
 &= 1 - p_1 + \sum_{n=3}^{+\infty} [p_1 \cdots p_{n-2} - p_1 \cdots p_{n-1}] \\
 &= 1 - \lim_{n \rightarrow \infty} (p_1 \cdots p_n), \quad (6)
 \end{aligned}$$

由此即得状态 0 常返 $\Leftrightarrow \lim_{n \rightarrow \infty} (p_1 p_2 \cdots p_n) = 0$.

(b) 由 (a) 中 (3), (4) 和 (5) 可知状态 0 的平均常返时为

$$\begin{aligned}\mu_0 &= \sum_{n=1}^{\infty} n f_{00}^{(n)} = 2q_1 + \sum_{n=3}^{+\infty} (np_1 \cdots p_{n-2} q_{n-1}) \\ &= 2(1 - p_1) + \sum_{n=3}^{+\infty} n[p_1 \cdots p_{n-2} - p_1 \cdots p_{n-1}]\end{aligned}$$

21. 分支过程 $\{X_n, n = 0, 1, 2, \cdots\}$ 中一个个体产生后代的分布为

$$\begin{aligned}P(Z_{ni} = 0) &= \frac{1}{8}, P(Z_{ni} = 1) = \frac{1}{2}, P(Z_{ni} = 2) = \frac{1}{4}, P(Z_{ni} = 3) = \frac{1}{8}, \\ i &= 1, \cdots, X_n, n = 0, 1, 2, \cdots, \quad (0)\end{aligned}$$

试求第 n 代总数 X_n 的均值和方差及群体消亡的概率.

解 由 (0) 可得第一代总数 X_1 的生成函数为

$$\begin{aligned}\phi_1(s) &= E(s^{Z_{01}}) = \sum_{k=0}^3 s^k P(Z_{01} = k) \\ &= \frac{1}{8}(1 + 4s + 2s^2 + s^3), \quad s \in (-\infty, +\infty),\end{aligned}$$

因而群体消亡的概率 π 满足

$$\frac{1}{8}(1 + 4\pi + 2\pi^2 + \pi^3) = \pi,$$

即

$$(\pi - 1)(\pi^2 + 3\pi - 1) = 0,$$

解之得 $\pi = \frac{\sqrt{13}-3}{2}$. 由 (0) 可得

$$\begin{aligned}\mu &= E(Z_{01}) = 0 \times \frac{1}{8} + 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + 3 \times \frac{1}{8} = \frac{11}{8}, \\ E(Z_{01}^2) &= 0^2 \times \frac{1}{8} + 1^2 \times \frac{1}{2} + 2^2 \times \frac{1}{4} + 3^2 \times \frac{1}{8} = \frac{21}{8}, \\ \sigma^2 &= Var(Z_{01}) = E(Z_{01}^2) - [E(Z_{01})]^2 = \frac{47}{64},\end{aligned}$$

因而, X_n 的均值为

$$\mu_X(n) = E(X_n) = \mu^n = \left(\frac{11}{8}\right)^n,$$

方差为

$$\begin{aligned}R_X(n, n) &= Var(X_n) = \sigma^2 \mu^{n-1} \frac{1 - \mu^n}{1 - \mu} \\ &= \frac{47}{24} \left[\left(\frac{11}{8}\right)^{2n-1} - \left(\frac{11}{8}\right)^{n-1} \right], \quad n = 0, 1, 2, \cdots.\end{aligned}$$

22. 若单一个体产生后代的分布为 $P(Z_{01} = 0) = q, P(Z_{01} = 1) = p, p + q = 1$, 并假定过程开始时的祖先数 $X_0 \equiv 1$, 试求分支过程 $\{X_n, n = 0, 1, 2, \dots\}$ 的第 n 代总数 X_n 的分布.

解 由题设可知第一代总数 X_1 的生成函数为

$$\phi_1(s) = E(s^{Z_{01}}) = s^0 P(Z_{01} = 0) + s^1 P(Z_{01} = 1) = q + ps, \quad s \in (-\infty, +\infty),$$

第二代总数 X_2 的生成函数为

$$\phi_2(s) = \phi_1(\phi_1(s)) = q + p(q + ps) = 1 - p^2 + p^2s, \quad s \in (-\infty, +\infty),$$

第三代总数 X_3 的生成函数为

$$\phi_3(s) = \phi_2(\phi_1(s)) = 1 - p^2 + p^2(q + ps) = 1 - p^3 + p^3s, \quad s \in (-\infty, +\infty),$$

.....

第 n 代总数 X_n 的生成函数为

$$\phi_n(s) = \phi_{n-1}(\phi_1(s)) = 1 - p^n + p^n s, \quad s \in (-\infty, +\infty).$$

这说明了 X_n 的分布为

$$P(X_n = 0) = 1 - p^n, \quad P(X_n = 1) = p^n, \quad n = 0, 1, 2, \dots$$

23. 一时齐连续时间 Markov 链有 0 和 1 两个状态, 在状态 0 和 1 的逗留时间服从参数为 $\lambda > 0$ 及 $\mu > 0$ 的指数分布. 试求在时刻 0 从状态 0 起始, t 时刻处于状态 0 的概率 $P_{00}(t)$.

解 以 T_0 表示从 $X(0) = 0$ 起始逗留状态 0 的时间, T_{2n+1} 表示从 $T_{2n} = t_{2n}$ 起始逗留状态 1 的时间, $n = 0, 1, 2, \dots$, T_{2n} 表示从 $T_{2n-1} = t_{2n-1}$ 起始逗留状态 0 的时间, $n = 1, 2, \dots$. 由题设可知, 给定 $X(0) = 0$ 时, T_0 的条件密度函数为

$$f_{T_0|X(0)}(t_0 | 0) = \lambda e^{-\lambda t_0}, \quad t_0 > 0,$$

给定 $(X(0), T_0) = (0, t_0)$ 时, T_1 的条件密度函数为

$$f_{T_1|(X(0), T_0)}(t_1 | (0, t_0)) = \mu e^{-\mu t_1}, \quad t_1 > 0,$$

给定 $(X(0), T_0, T_1) = (0, t_0, t_1)$ 时, T_2 的条件密度函数为

$$f_{T_2|(X(0), T_0, T_1)}(t_2 | (0, t_0, t_1)) = \lambda e^{-\lambda t_2}, \quad t_2 > 0,$$

.....

给定 $(X(0), T_0, T_1, T_2, \dots, T_{2n}) = (0, t_0, t_1, t_2, \dots, t_{2n})$ 时, T_{2n+1} 的条件密度函数为

$$f_{T_{2n+1}|(X(0), T_0, T_1, T_2, \dots, T_{2n})}(t_{2n+1} | (0, t_0, t_1, t_2, \dots, t_{2n})) = \mu e^{-\mu t_{2n+1}}, \quad t_{2n+1} > 0,$$

$$n = 0, 1, 2, \dots,$$

给定 $(X(0), T_0, T_1, T_2, \dots, T_{2n-1}) = (0, t_0, t_1, t_2, \dots, t_{2n-1})$ 时, T_{2n} 的条件密度函数为

$$f_{T_{2n}|(X(0), T_0, T_1, T_2, \dots, T_{2n-1})}(t_{2n} | (0, t_0, t_1, t_2, \dots, t_{2n-1})) = \lambda e^{-\lambda t_{2n}}, \quad t_{2n} > 0, \\ n = 1, 2, \dots,$$

因而, 给定 $X(0) = 0$ 时, 随机向量 $(T_0, T_1, T_2, \dots, T_{2n})$ 的条件密度函数为

$$\begin{aligned} & f_{(T_0, T_1, T_2, \dots, T_{2n})|X(0)}(t_0, t_1, t_2, \dots, t_{2n} | 0) \\ &= f_{T_0|X(0)}(t_0 | 0) f_{T_1|X(0), T_0}(t_1 | (0, t_0)) f_{T_2|X(0), T_0, T_1}(t_2 | (0, t_0, t_1)) \cdots \\ & \quad \cdot f_{T_{2n}|X(0), T_0, T_1, T_2, \dots, T_{2n-1}}(t_{2n} | (0, t_0, t_1, t_2, \dots, t_{2n-1})) \\ &= \lambda^{n+1} \mu^n e^{-\lambda \sum_{k=0}^n t_{2k}} e^{-\mu \sum_{k=1}^n t_{2k-1}}, \quad t_0, t_1, t_2, \dots, t_{2n} > 0, n = 0, 1, 2, \dots, \end{aligned} \quad (1)$$

给定 $X(0) = 0$ 时, 随机向量 $(T_0, T_1, T_2, \dots, T_{2n-1})$ 的条件密度函数为

$$\begin{aligned} & f_{(T_0, T_1, T_2, \dots, T_{2n-1})|X(0)}(t_0, t_1, t_2, \dots, t_{2n-1} | 0) \\ &= f_{T_0|X(0)}(t_0 | 0) f_{T_1|X(0), T_0}(t_1 | (0, t_0)) f_{T_2|X(0), T_0, T_1}(t_2 | (0, t_0, t_1)) \cdots \\ & \quad \cdot f_{T_{2n-1}|X(0), T_0, T_1, T_2, \dots, T_{2n-2}}(t_{2n-1} | (0, t_0, t_1, t_2, \dots, t_{2n-2})) \\ &= \lambda^n \mu^n e^{-\lambda \sum_{k=0}^{n-1} t_{2k}} e^{-\mu \sum_{k=1}^n t_{2k-1}}, \quad t_0, t_1, t_2, \dots, t_{2n-1} > 0, n = 1, 2, \dots. \end{aligned} \quad (2)$$

(1) 和 (2) 表明了, 给定 $X(0) = 0$ 时, $T_0, T_1, T_2, \dots$, 互相条件独立, 且

$$T_{2n} | X(0) = 0 \sim P(\lambda), \quad T_{2n+1} | X(0) = 0 \sim P(\mu), \quad n = 0, 1, 2, \dots, \quad (3)$$

其中 $P(\lambda)$ 表示参数为 λ 的指数分布. 若记

$$W_n = \sum_{k=0}^n T_k, \quad n = 0, 1, 2, \dots, \quad (4)$$

则从 $X(0) = 0$ 起始于时刻 t 处处于状态 0 的事件为

$$\{X(0) = 0, X(t) = 0\} = \{W_0 > t\} + \sum_{n=1}^{+\infty} \{W_{2n} > t, W_{2n-1} \leq t\},$$

因此

$$\begin{aligned} P_{00}(t) &= P(X(t) = 0 | X(0) = 0) \\ &= P(W_0 > t | X(0) = 0) + \sum_{n=1}^{+\infty} P(W_{2n} > t, W_{2n-1} \leq t | X(0) = 0) \\ &= P(W_0 > t | X(0) = 0) \\ & \quad + \sum_{n=1}^{+\infty} (P(W_{2n} > t | X(0) = 0) - P(W_{2n-1} > t | X(0) = 0)), \quad t > 0. \end{aligned} \quad (5)$$

若记

$$U_n = \sum_{k=0}^n T_{2k}, \quad n = 0, 1, 2, \dots, \quad V_n = \sum_{k=1}^n T_{2k-1}, \quad n = 1, 2, \dots, \quad (6)$$

则由 (4) 得

$$W_{2n} = U_n + V_n, \quad W_{2n-1} = U_{n-1} + V_n, \quad n = 1, 2, \dots, \quad (7)$$

而由 (3) 可知, 给定 $X(0) = 0$ 时, U_n 与 V_n 条件独立, 且

$$f_{U_n|X(0)=0}(u | 0) = \frac{\lambda^{n+1}}{n!} u^n e^{-\lambda u}, \quad u > 0, n = 0, 1, 2, \dots, \quad (8)$$

$$f_{V_n|X(0)=0}(v | 0) = \frac{\mu^n}{(n-1)!} v^{n-1} e^{-\mu v}, \quad v > 0, n = 1, 2, \dots. \quad (9)$$

故

$$\begin{aligned} & P(W_{2n} > t | X(0) = 0) \\ &= P(U_n + V_n > t | X(0) = 0) \\ &= 1 - \frac{\lambda^{n+1} \mu^n}{n!(n-1)!} \iint_{u,v>0, u+v \leq t} u^n v^{n-1} e^{-(\lambda u + \mu v)} du dv \\ &= 1 - \frac{\lambda^{n+1} \mu^n}{n!(n-1)!} \int_0^t v^{n-1} e^{-\mu v} dv \int_0^{t-v} u^n e^{-\lambda u} du, \quad n = 1, 2, \dots, \end{aligned}$$

$$\begin{aligned} & P(W_{2n-1} > t | X(0) = 0) \\ &= P(U_{n-1} + V_n > t | X(0) = 0) \\ &= 1 - \frac{(\lambda \mu)^n}{[(n-1)!]^2} \iint_{u,v>0, u+v \leq t} (uv)^{n-1} e^{-(\lambda u + \mu v)} du dv \\ &= 1 - \frac{(\lambda \mu)^n}{[(n-1)!]^2} \int_0^t v^{n-1} e^{-\mu v} dv \int_0^{t-v} u^{n-1} e^{-\lambda u} du, \quad n = 1, 2, \dots, \end{aligned}$$

因而

$$\begin{aligned} & P(W_{2n} > t | X(0) = 0) - P(W_{2n-1} > t | X(0) = 0) \\ &= \frac{(\lambda \mu)^n}{[(n-1)!]^2} \int_0^t v^{n-1} e^{-\mu v} dv \int_0^{t-v} u^{n-1} e^{-\lambda u} du \\ &\quad - \frac{\lambda^{n+1} \mu^n}{n!(n-1)!} \int_0^t v^{n-1} e^{-\mu v} dv \int_0^{t-v} u^n e^{-\lambda u} du. \end{aligned} \quad (10)$$

而

$$\begin{aligned} \int_0^{t-v} u^n e^{-\lambda u} du &= -\frac{1}{\lambda} \int_0^{t-v} u^n d e^{-\lambda u} \\ &= \frac{n}{\lambda} \int_0^{t-v} u^{n-1} e^{-\lambda u} du - \frac{1}{\lambda} (t-v)^n e^{-\lambda(t-v)}, \quad n = 1, 2, \dots, \end{aligned}$$

因而由 (10) 得

$$\begin{aligned} & P(W_{2n} > t \mid X(0) = 0) - P(W_{2n-1} > t \mid X(0) = 0) \\ &= \frac{(\lambda\mu)^n}{n!(n-1)!} \int_0^t v^{n-1}(t-v)^n e^{-[\mu v + \lambda(t-v)]} dv, \quad n = 1, 2, \dots \end{aligned} \quad (11)$$

将其代入 (5) 中得

$$P_{00}(t) = e^{-\lambda t} + \sum_{n=1}^{+\infty} \frac{(\lambda\mu)^n}{n!(n-1)!} \int_0^t v^{n-1}(t-v)^n e^{-[\mu v + \lambda(t-v)]} dv, \quad t > 0. \quad (12)$$

当 $\lambda = \mu$ 时, 由上式得

$$\begin{aligned} P_{00}(t) &= e^{-\lambda t} \left(1 + \sum_{n=1}^{+\infty} \frac{\lambda^{2n}}{n!(n-1)!} \int_0^t v^{n-1}(t-v)^n dv \right) \\ &= e^{-\lambda t} \sum_{n=0}^{+\infty} \frac{(\lambda t)^{2n}}{(2n)!} = \frac{1 + e^{-2\lambda t}}{2}, \quad t > 0. \end{aligned} \quad (13)$$

当 $\lambda \neq \mu$ 时, 由 (12) 得

24. 在第 23 题中, 定义 $N(t)$ 为过程在 $[0, t]$ 中改变状态的次数, 试求 $N(t)$ 的概率分布.

解 沿用上题题解中的记号有

$$\begin{aligned} P(N(t) = 0) &= P(W_0 > t \mid X(0) = 0) \\ &= P(T_0 > t \mid X(0) = 0) = e^{-\lambda t}, \quad t > 0, \end{aligned} \quad (1)$$

$$\begin{aligned} P(N(t) = 1) &= P(W_0 \leq t, W_1 > t \mid X(0) = 0) \\ &= P(W_1 > t \mid X(0) = 0) - P(W_0 > t \mid X(0) = 0) \\ &= 1 - e^{-\lambda t} - \lambda\mu \int_0^t e^{-\mu v} dv \int_0^{t-v} e^{-\lambda u} du \\ &= e^{-\mu t} - e^{-\lambda t} + \mu \int_0^t e^{-[\mu v + \lambda(t-v)]} dv \\ &= \begin{cases} \lambda t e^{-\lambda t}, & \lambda = \mu, \\ \frac{\lambda}{\lambda - \mu} (e^{-\mu t} - e^{-\lambda t}), & \lambda \neq \mu, \end{cases} \quad t > 0, \end{aligned} \quad (2)$$

$$\begin{aligned} P(N(t) = 2k) &= P(W_{2k-1} \leq t, W_{2k} > t \mid X(0) = 0) \\ &= P(W_{2k} > t \mid X(0) = 0) - P(W_{2k-1} > t \mid X(0) = 0) \\ &= \frac{(\lambda\mu)^k}{k!(k-1)!} \int_0^t v^{k-1}(t-v)^k e^{-[\mu v + \lambda(t-v)]} dv, \\ & \quad k = 1, 2, \dots, \end{aligned} \quad (3)$$

$$P(N(t) = 2k+1) = P(W_{2k} \leq t, W_{2k+1} > t \mid X(0) = 0)$$

$$\begin{aligned}
&= P(W_{2k+1} > t \mid X(0) = 0) - P(W_{2k} > t \mid X(0) = 0) \\
&= \frac{\lambda^{k+1} \mu^k}{k!(k-1)!} \iint_{u,v>0, u+v \leq t} u^k v^{k-1} e^{-(\lambda u + \mu v)} du dv \\
&\quad - \frac{(\lambda \mu)^{k+1}}{(k!)^2} \iint_{u,v>0, u+v \leq t} (uv)^k e^{-(\lambda u + \mu v)} du dv \\
&= \frac{\lambda^{k+1} \mu^k}{k!(k-1)!} \int_0^t u^k e^{-\lambda u} du \int_0^{t-u} v^{k-1} e^{-\mu v} dv \\
&\quad - \frac{(\lambda \mu)^{k+1}}{(k!)^2} \int_0^t u^k e^{-\lambda u} du \int_0^{t-u} v^k e^{-\mu v} dv \\
&= \frac{\lambda^{k+1} \mu^k}{(k!)^2} \int_0^t u^k (t-u)^k e^{-[\lambda u + \mu(t-u)]} du, \\
&\hspace{25em} k = 1, 2, \dots \quad (4)
\end{aligned}$$

当 $\lambda = \mu$ 时, 由 (3) 和 (4) 可得

$$\begin{aligned}
P(N(t) = 2k) &= P(W_{2k-1} \leq t, W_{2k} > t \mid X(0) = 0) \\
&= \frac{(\lambda)^{2k}}{k!(k-1)!} e^{-\lambda t} \int_0^t v^{k-1} (t-v)^k dv \\
&= \frac{(\lambda t)^{2k}}{(2k)!} e^{-\lambda t}, \quad k = 1, 2, \dots, \quad (5)
\end{aligned}$$

$$\begin{aligned}
P(N(t) = 2k+1) &= P(W_{2k} \leq t, W_{2k+1} > t \mid X(0) = 0) \\
&= \frac{\lambda^{2k+1}}{(k!)^2} e^{-\lambda t} \int_0^t u^k (t-u)^k du \\
&= \frac{(\lambda t)^{2k+1}}{(2k+1)!} e^{-\lambda t}, \quad k = 1, 2, \dots, \quad (6)
\end{aligned}$$

(1),(2),(5) 和 (6) 说明了, 当 $\lambda = \mu$ 时, $N(t)$ 服从参数为 λt 的 Poisson 分布, $t > 0$. ■

习题 4

5. 设 $\{X_n, n = 1, 2, \dots\}$ 是一独立同分布随机变量序列, $P(X_1 = 1) = p$, $P(X_1 = -1) = q, p + q = 1$. 令 $S_n = \frac{X_1 + \dots + X_n}{\sqrt{n}}, n = 1, 2, \dots$. 试求序列 $\{S_n, n = 1, 2, \dots\}$ 的自协方差函数和自相关函数, 并证明 $\{S_n, n = 1, 2, \dots\}$ 不平稳.

解证 由题设可知

$$E(X_n) = 1 \cdot P(X_n = 1) + (-1) \cdot P(X_n = -1) = p - q, \quad n = 1, 2, \dots, \quad (1)$$

$$E(X_n^2) = 1^2 \cdot P(X_n = 1) + (-1)^2 \cdot P(X_n = -1) = p + q = 1, \quad n = 1, 2, \dots,$$

$$\text{Var}(X_n) = E(X_n^2) - [E(X_n)]^2 = 1 - (p - q)^2, \quad n = 1, 2, \dots. \quad (2)$$

因而, $\{S_n, n = 1, 2, \dots\}$ 的均值函数为

$$m_S(n) = \frac{1}{\sqrt{n}} \sum_{k=1}^n E(X_k) = \sqrt{n}(p - q), \quad n = 1, 2, \dots, \quad (3)$$

自协方差函数为

$$\begin{aligned} R_S(m, n) &= \text{Cov} \left(\frac{1}{\sqrt{m}} \sum_{k=1}^m E(X_k), \frac{1}{\sqrt{n}} \sum_{l=1}^n E(X_l) \right) = \frac{1}{\sqrt{mn}} \sum_{k=1}^m \sum_{l=1}^n \text{Cov}(X_k, X_l) \\ &= \frac{1}{\sqrt{mn}} \sum_{k=1}^{\min\{m, n\}} \text{Var}(X_k) = \frac{\min\{m, n\}}{\sqrt{mn}} [1 - (p - q)^2], \quad m, n = 1, 2, \dots. \end{aligned} \quad (4)$$

由 (3) 可知, 欲使 $\{S_n, n = 1, 2, \dots\}$ 平稳, 必须 $p = q = \frac{1}{2}$. 而当 $p = q = \frac{1}{2}$ 时, 由 (4) 得

$$R_S(m, n) = \frac{\min\{m, n\}}{\sqrt{mn}}, \quad m, n = 1, 2, \dots.$$

这说明了 $R_S(m, n)$ 不可能只与 $m - n$ 有关, 故 $\{S_n, n = 1, 2, \dots\}$ 不平稳. ■

6. 设 $\{X(t), t \in (-\infty, +\infty)\}$ 平稳, 对每一 $t \in (-\infty, +\infty)$, $X'(t)$ 存在. 证明对每一 $t \in (-\infty, +\infty)$, $X(t)$ 与 $X'(t)$ 不相关.

证

10. 设 $\{X(t), t \in (-\infty, +\infty)\}$ 是一复值平稳过程, 证明

$$E[|X(t + \tau) - X(t)|^2] = 2\text{Re}(R(0) - R(\tau)).$$

证 由 $\{X(t), t \in (-\infty, +\infty)\}$ 的平稳性知

$$\begin{aligned} E[|X(t + \tau) - X(t)|^2] &= E[|(X(t + \tau) - m) - (X(t) - m)|^2] \\ &= E[|X(t + \tau) - m|^2] + E[|X(t) - m|^2] \\ &\quad - E[(X(t + \tau) - m)\overline{(X(t) - m)}] - E[(X(t) - m)\overline{(X(t + \tau) - m)}] \end{aligned}$$

$$=2R(0) - R(-\tau) - R(\tau),$$

其中 $m = E(X(t)), t \in (-\infty, +\infty)$. 而 $R(-\tau) = \overline{R(\tau)}$, 因而由上式可得

$$\begin{aligned} E[|X(t+\tau) - X(t)|^2] &= 2R(0) - R(\tau) - \overline{R(\tau)} \\ &= 2\operatorname{Re}(R(0) - R(\tau)), \quad t \in (-\infty, +\infty). \end{aligned}$$

■

11. 设 $\{X(t), t \in (-\infty, +\infty)\}$ 是一平稳 Gauss 过程, 自协方差函数为 $R(\tau)$. 证明

$$P(X'(t) \leq a) = \Phi\left(\frac{a}{\sqrt{-R''(0)}}\right), \quad a \in (-\infty, +\infty),$$

其中 $\Phi(\cdot)$ 为标准正态分布函数.

证 由题设可知

$$\begin{pmatrix} X(t) \\ X(t+h) \end{pmatrix} \sim N_2\left(\begin{pmatrix} \mu \\ \mu \end{pmatrix}, \begin{pmatrix} R(0) & R(h) \\ R(h) & R(0) \end{pmatrix}\right), \quad \forall t, t+h,$$

由此可得

$$X(t+h) - X(t) \sim N(0, 2(R(0) - R(h))), \quad \forall t, t+h.$$

因而

$$P\left(\frac{X(t+h) - X(t)}{h} \leq a\right) = \Phi\left(\frac{ah}{\sqrt{2(R(0) - R(h))}}\right), \quad \forall t, h > 0.$$

故

$$\begin{aligned} P(X'(t) \leq a) &= \lim_{h \rightarrow 0+} P\left(\frac{X(t+h) - X(t)}{h} \leq a\right) \\ &= \lim_{h \rightarrow 0+} \Phi\left(\frac{ah}{\sqrt{2(R(0) - R(h))}}\right) \\ &= \Phi\left(\frac{a}{\sqrt{-R''(0)}}\right), \quad \forall t. \end{aligned}$$

■

§0.1 ?

输入定理和公式的例子

定理0.1.1 ([1, Theorem I.4.3]). 设 $f \in C^1(X, \mathbf{R})$ 满足条件 (C), 则我们有

$$\sum_{q=0}^{\infty} M_q t^q = \sum_{q=0}^{\infty} \beta_q t^q + (1+t) Q(t), \quad (0.1.1)$$

这里 Q 是有非负整系数的形式级数,.

引理、命题等可类似输入。

引用时用交叉引用: 定理 0.1.1 中的式子 (0.1.1) 称为 Morse 不等式.

输入多行公式, 用 align 环境:

$$\begin{aligned} \operatorname{ind}(\nabla \varphi, v) &= \sum_{q=0}^{\infty} (-1)^q \operatorname{rank} C_q(\varphi, v) \\ &= \sum_{q=0}^{\infty} (-1)^q \operatorname{rank} C_q(f, v + \psi(v)) \\ &= \operatorname{ind}(\nabla f, v + \psi(v)). \end{aligned} \quad (0.1.2)$$

若不想编号就用 align* 环境, 这时不需要写 nonumber 命令. align 比 eqnarray 的好处在于, 等号或不等号两边不会留太多空白.

参考文献

- [1] K. C. CHANG, Infinite dimensional Morse theory and multiple solution problem, Birkhäuser, Bostom, 1993.
- [2] 其他文献同样添加。

作者简介

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致谢

致谢的内容

某某某

2003 年 3 月
于中国 XXX 研究所